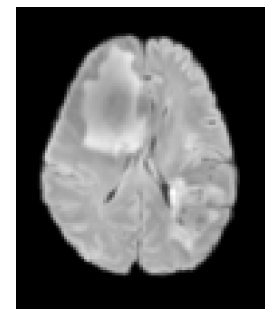


Deep Learning for Medical Imaging

Uncertainties and Explainability

Valentine Wagnier-Dauchelle
CREATIS INSA Lyon

Why explain? Why quantify uncertainty?



Input data

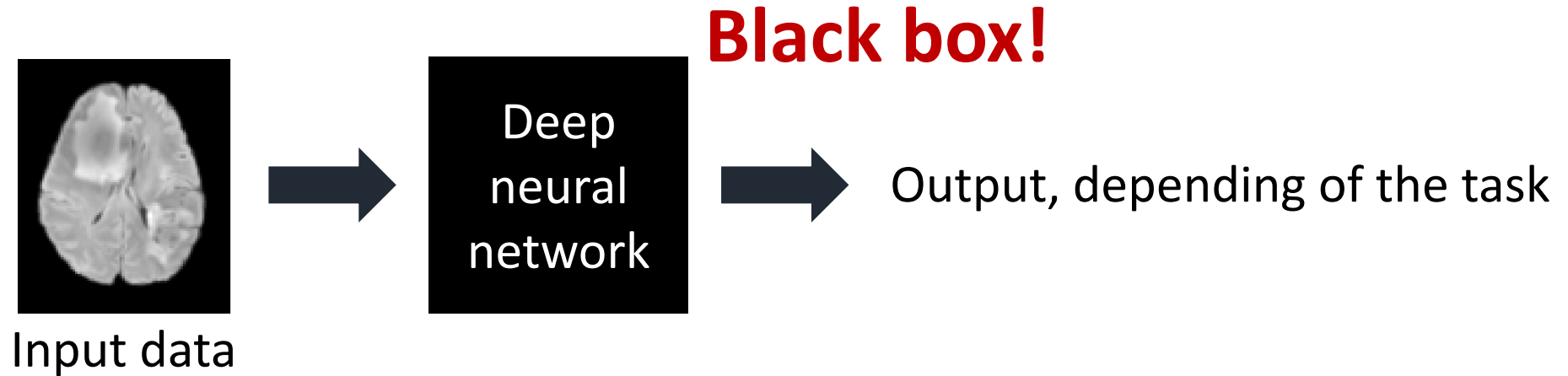


Deep
neural
network

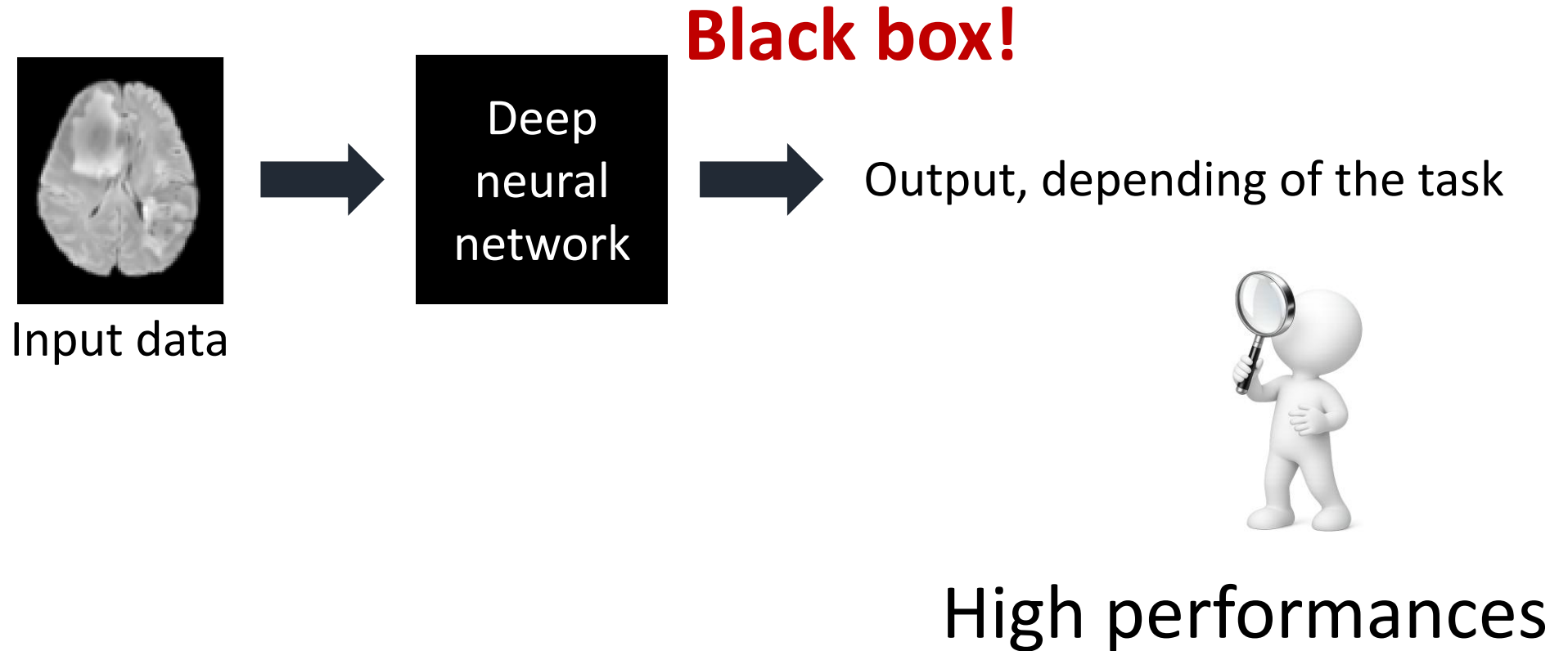


Output, depending of the task

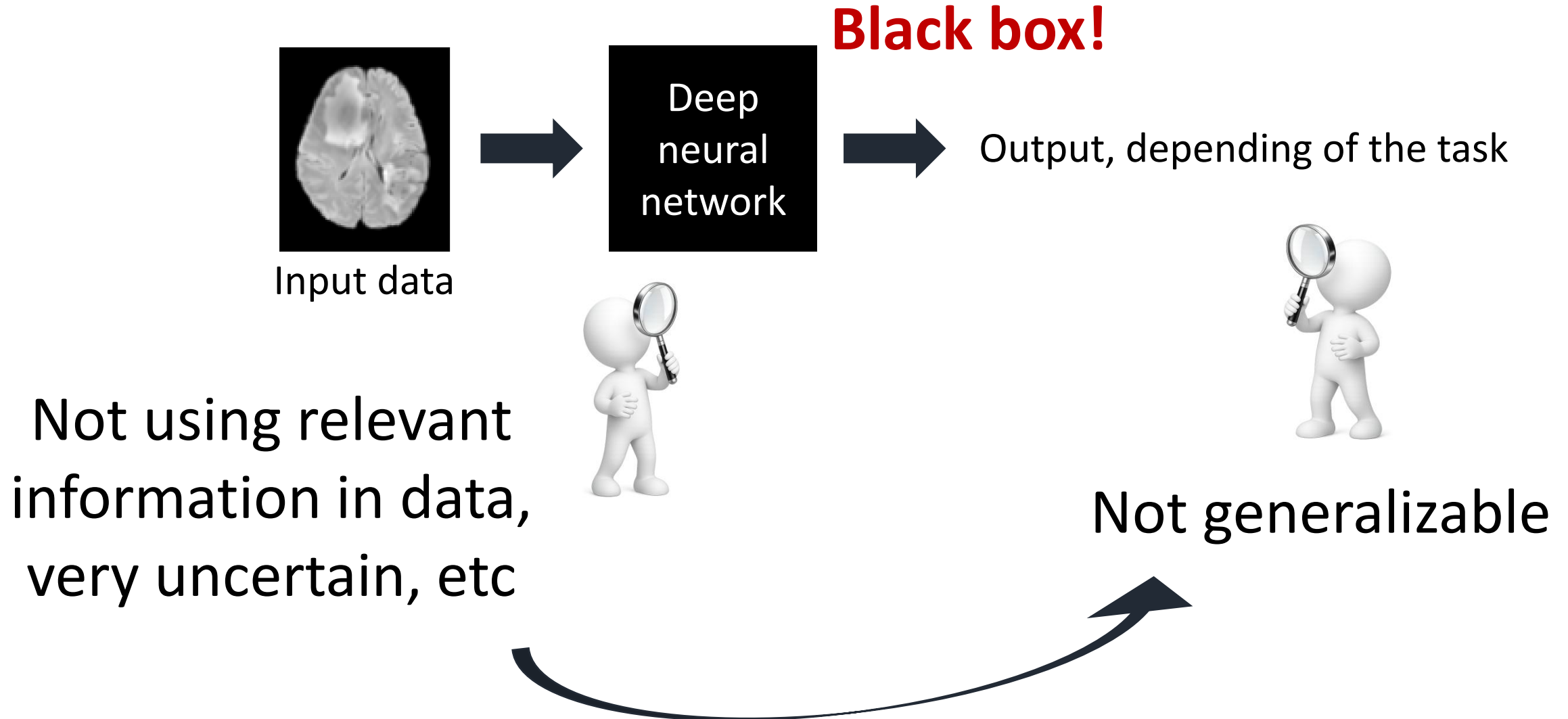
Why explain? Why quantify uncertainty?



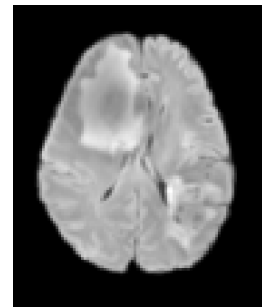
Why explain? Why quantify uncertainty?



Why explain? Why quantify uncertainty?



Why explain? Why quantify uncertainty?



Input data



Deep
neural
network

Black box!



Output, depending of the task



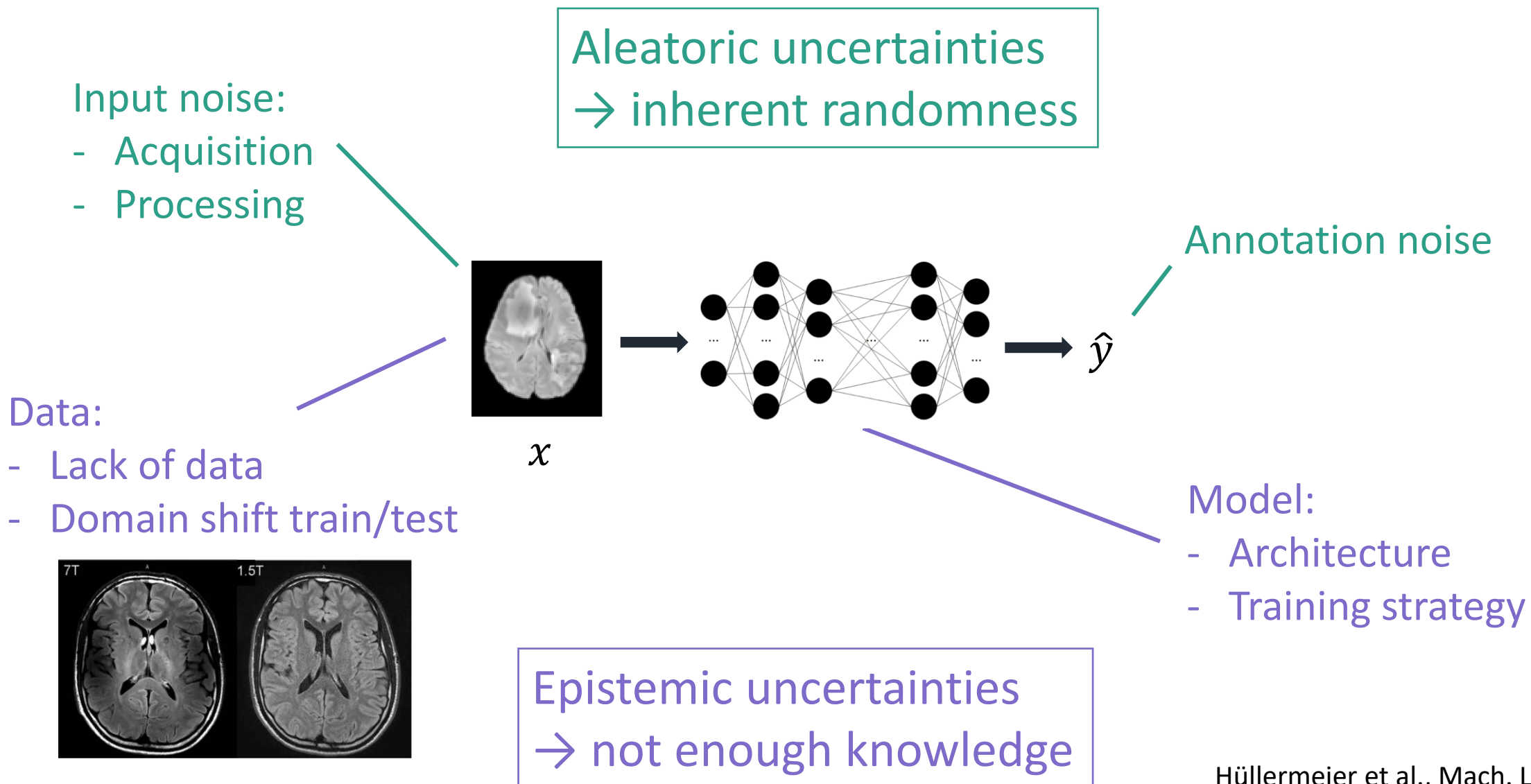
Not using relevant
information in data,
very uncertain, etc



Not generalizable

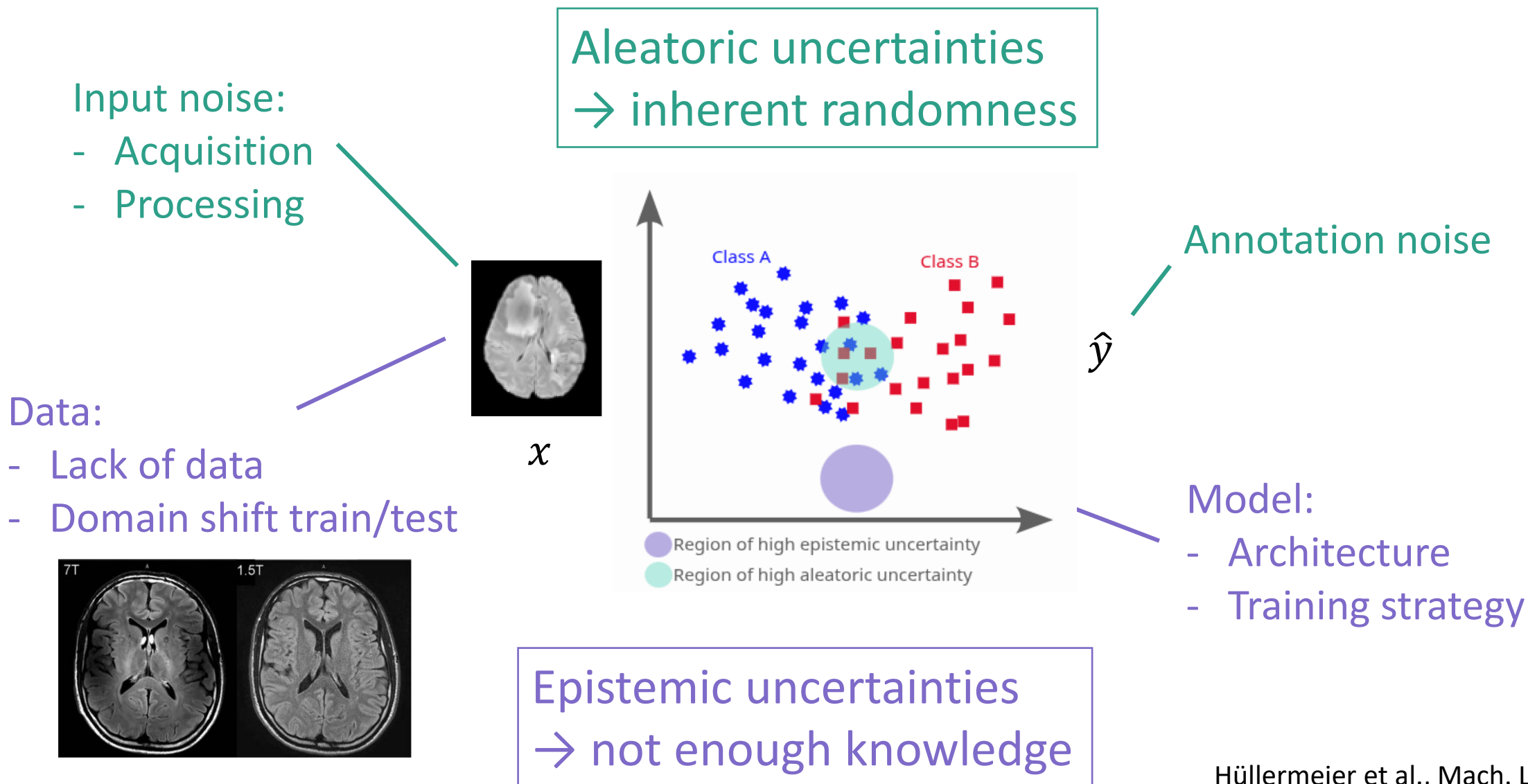
In medical context: need safeguard!

Source of uncertainties/errors



Hüllermeier et al., Mach. Learning, 2021

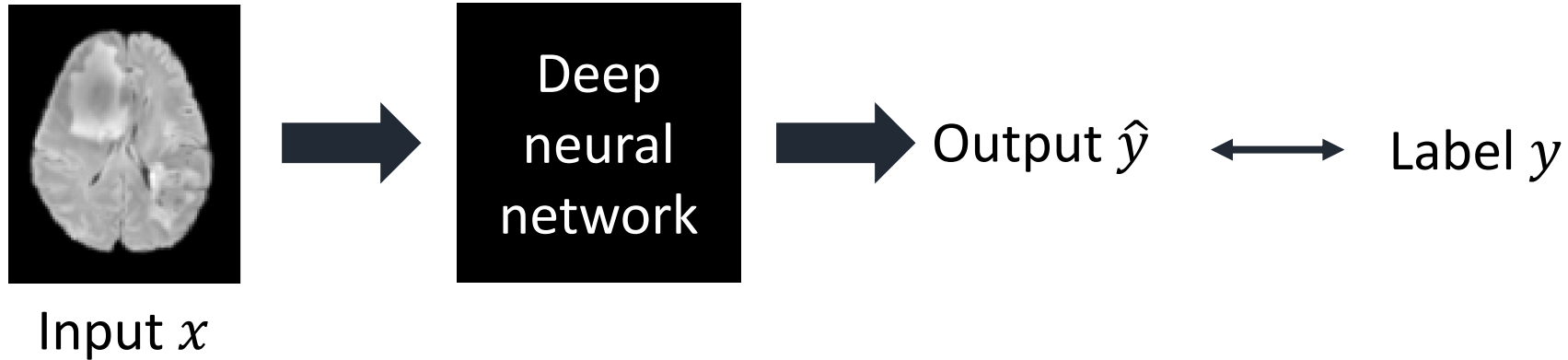
Source of uncertainties/errors



Hüllermeier et al., Mach. Learning, 2021

Domain shift

Difference between the train and test distribution



Covariate shift: Input distribution

$$P_{train}(X) \neq P_{test}(X)$$
$$P_{train}(Y|X) = P_{test}(Y|X)$$

Ex: Acquisition, demographics

Label shift: class frequency

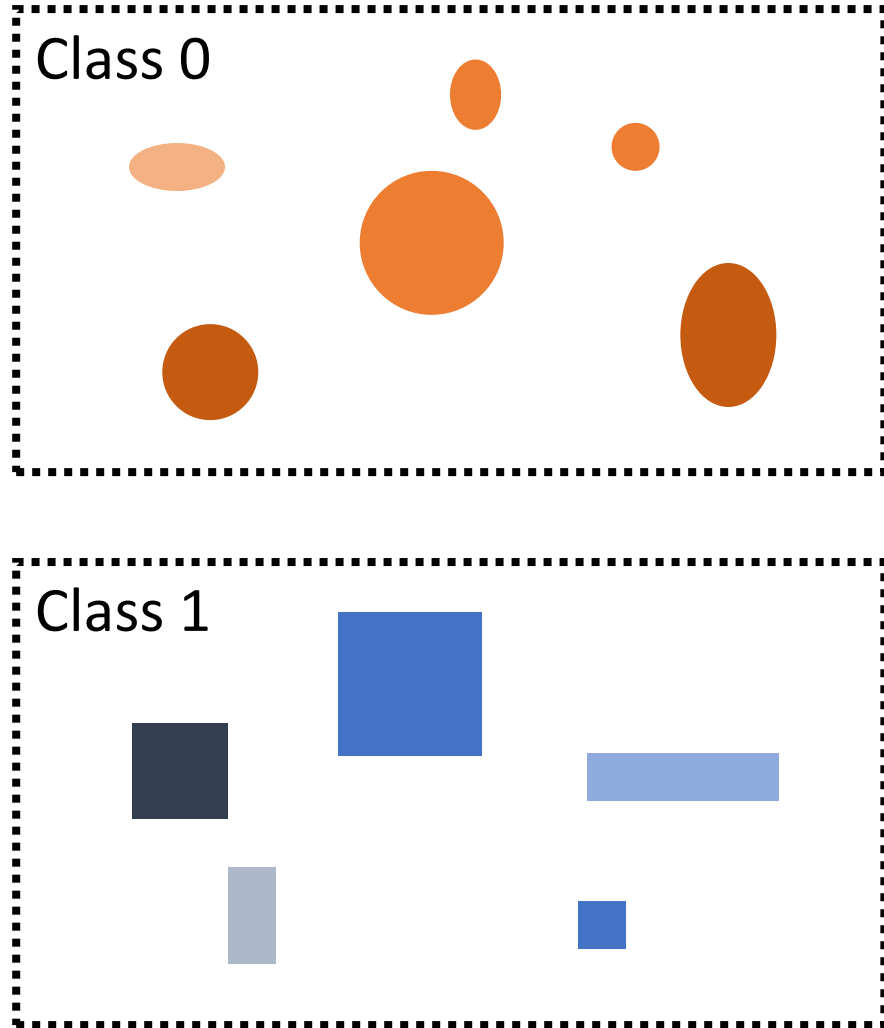
$$P_{train}(Y) \neq P_{test}(Y)$$
$$P_{train}(Y|X) = P_{test}(Y|X)$$

Concept shift : relation between input and label

$$P_{train}(Y|X) \neq P_{test}(Y|X)$$
$$P_{train}(X) = P_{test}(X)$$

Ex: Clinical guideline, bias

Dataset biases, confounder



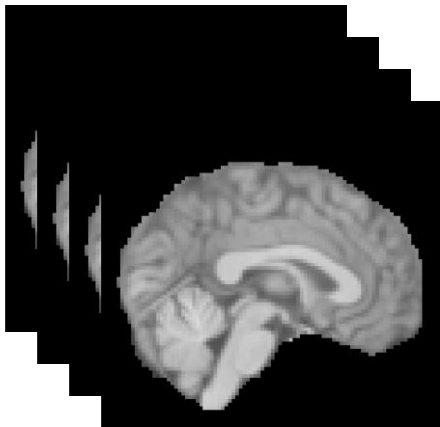
Classifier

What is the shape?

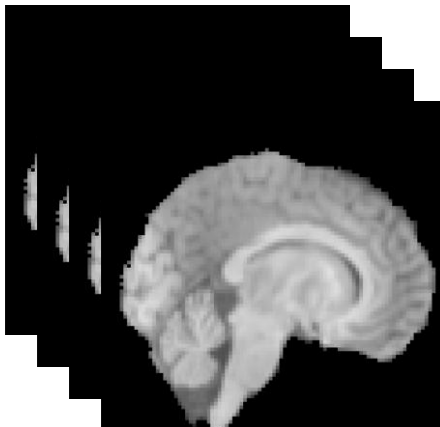
Confounder: Color

Dataset biases, confounder

Class 0



Class 1



Classifier

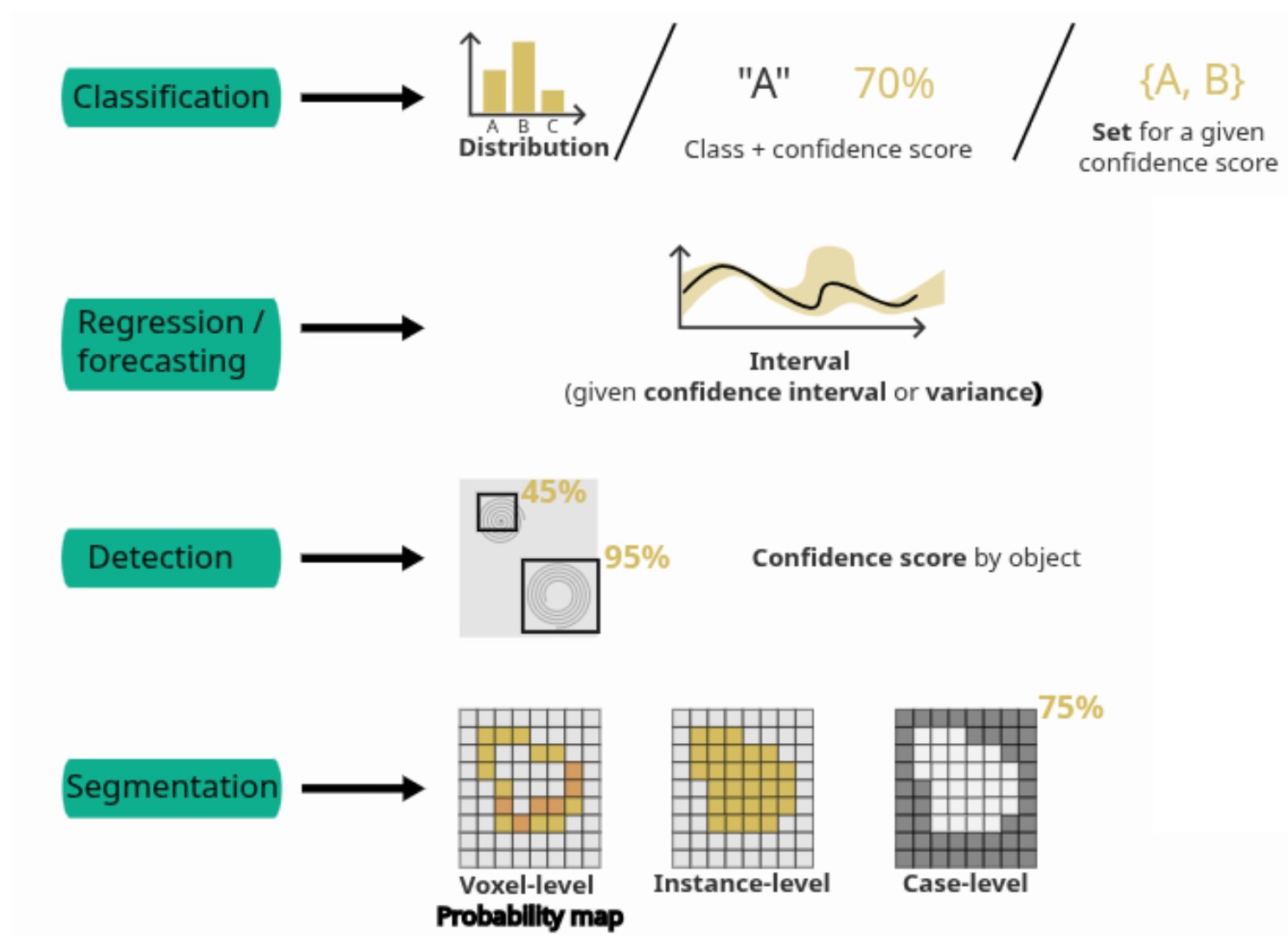
Which class?

Confounders:

- Imaging parameters: scanner brand, acquisition parameters, ...
- Population : age, morphology, ...
- Something else ...

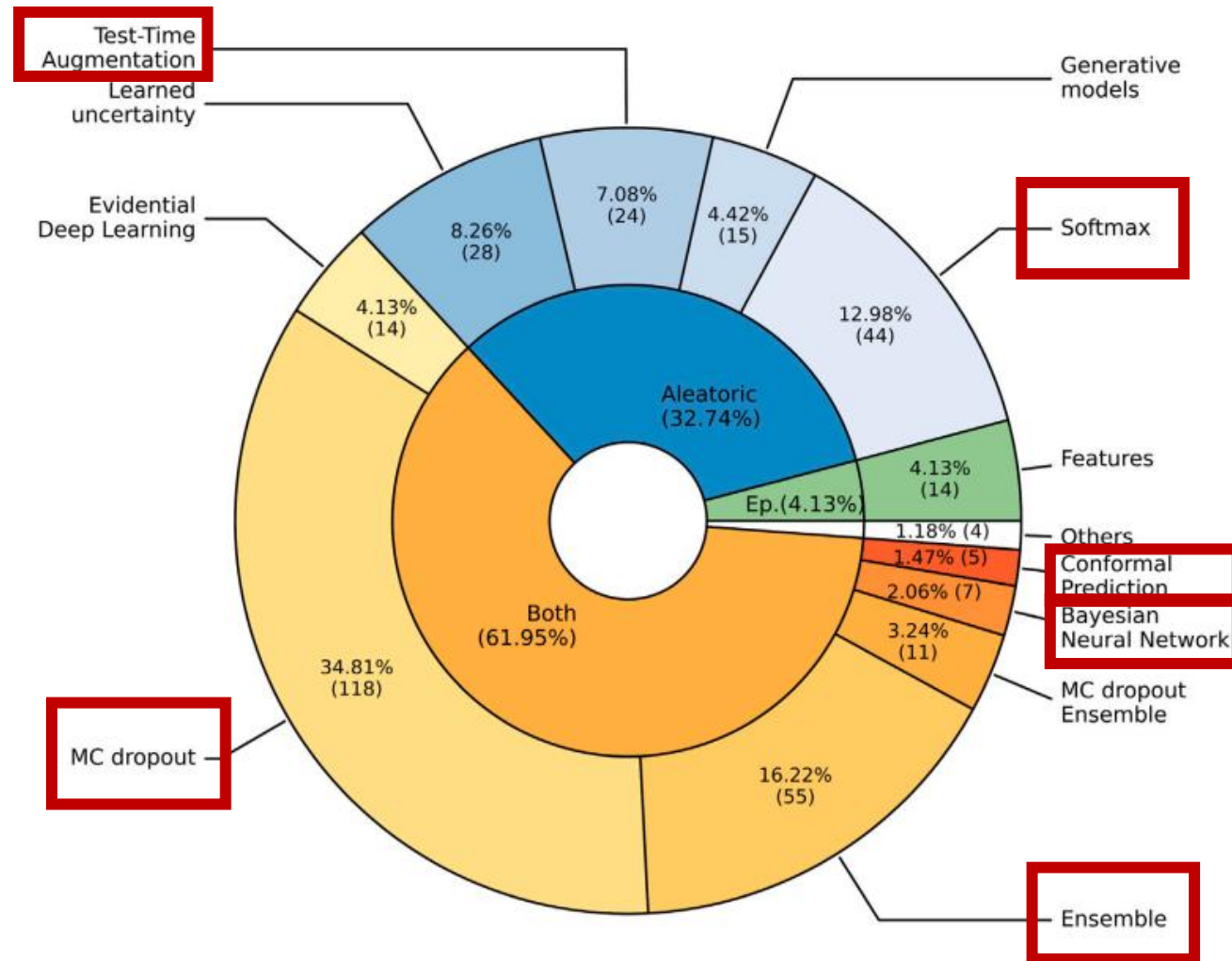
Uncertainty estimation

Different task = different uncertainties



<https://creatis-myriad.github.io/tutorials/2026-03-19-tuto-UQ-classif.html>

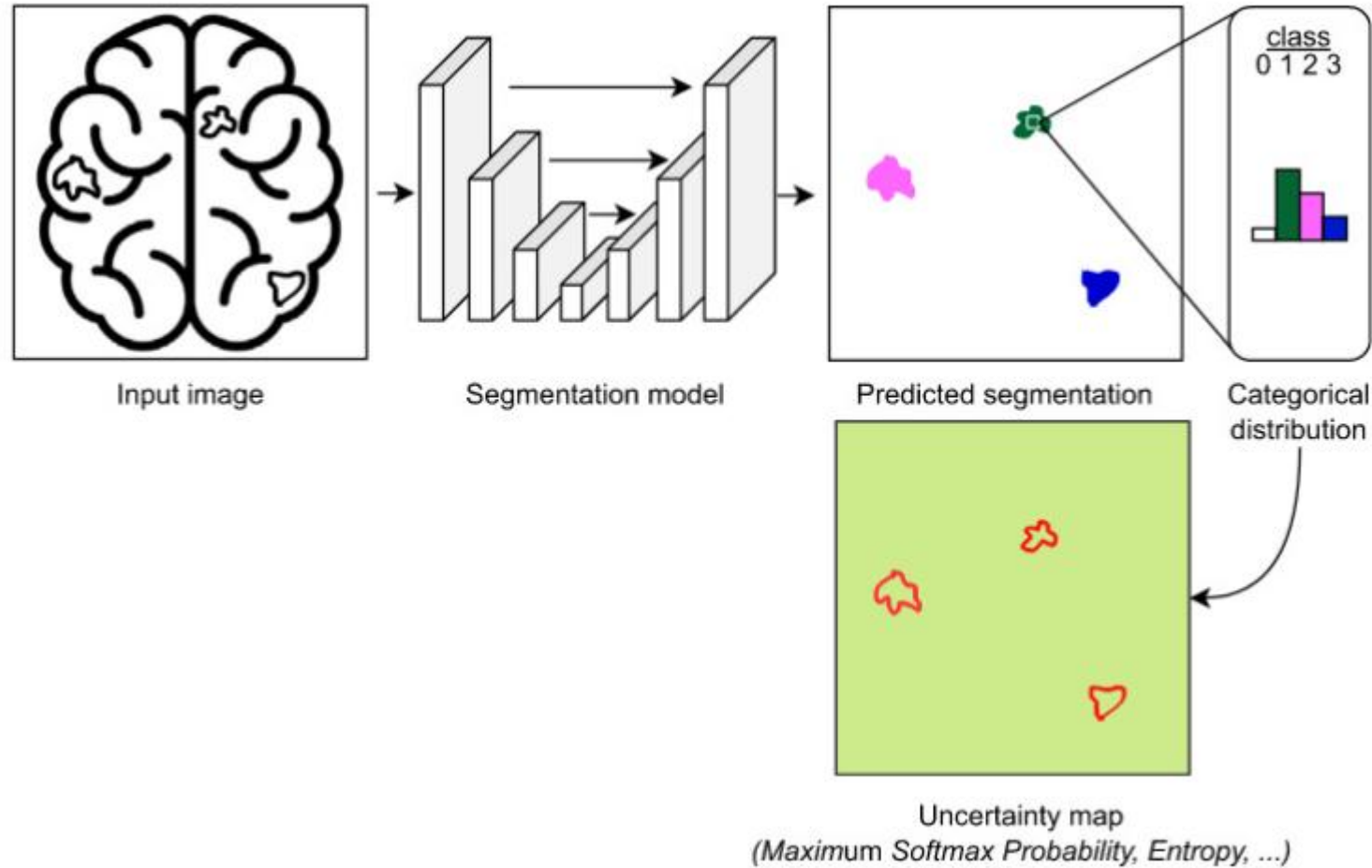
Different methods



Lambert et al., Artificial Intelligence for Medicine, 2024

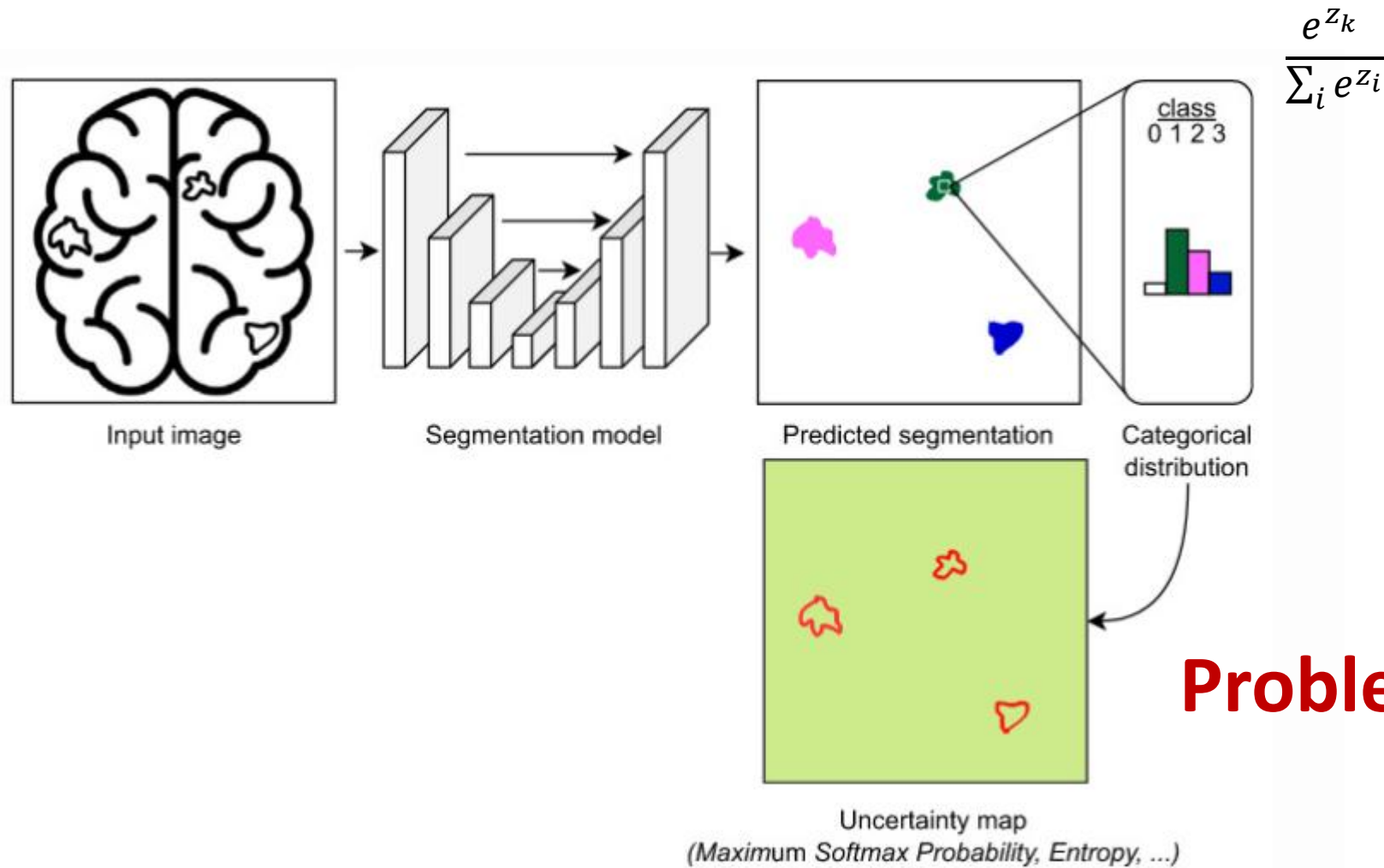
Quantification: softmax

$$\frac{e^{z_k}}{\sum_i e^{z_i}}$$



Lambert et al., Artificial Intelligence for Medicine, 2024

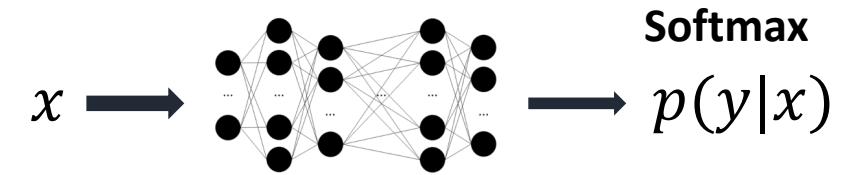
Quantification: softmax



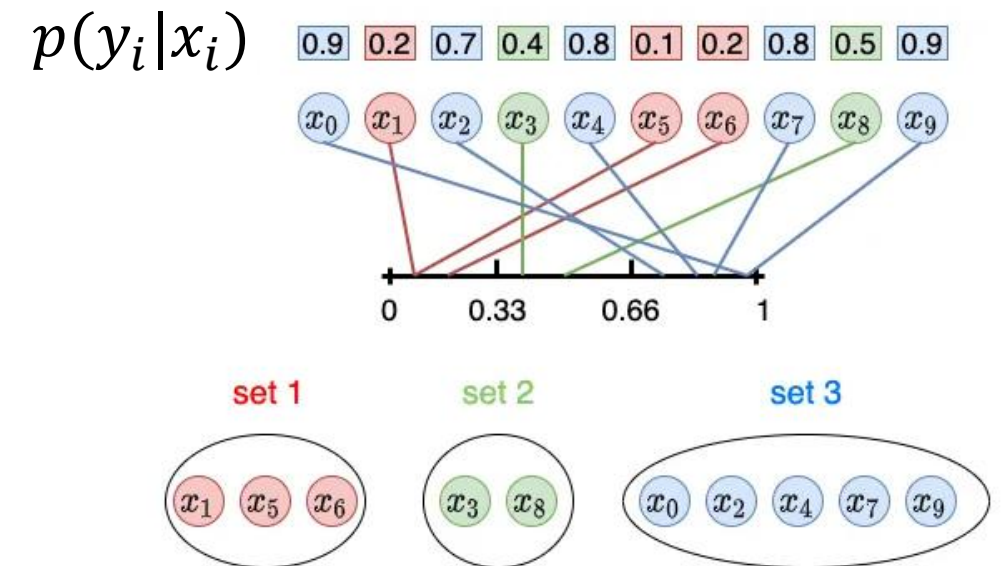
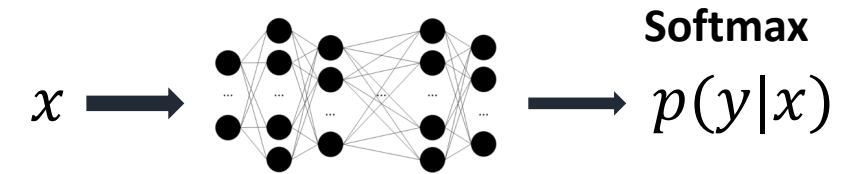
Problem: not calibrated !

Lambert et al., Artificial Intelligence for Medicine, 2024

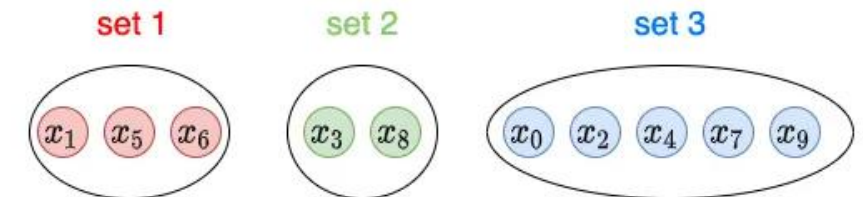
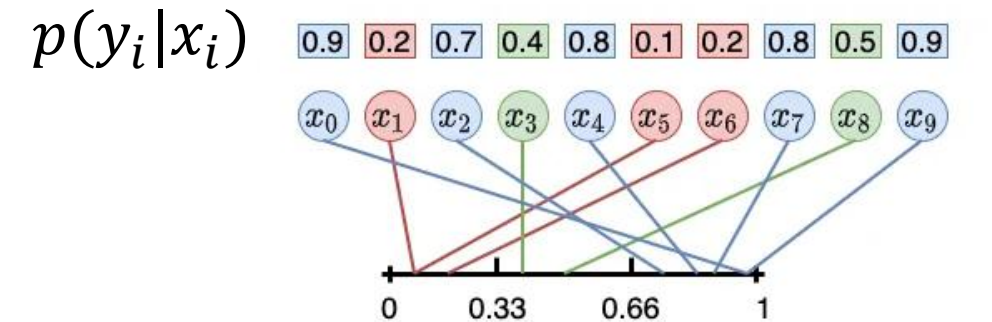
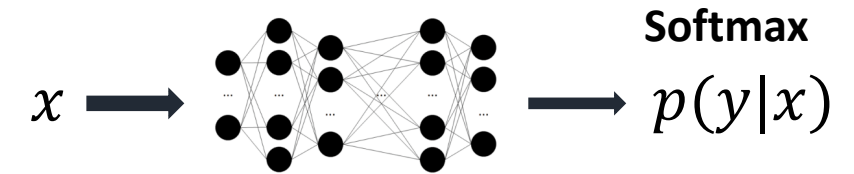
Calibration: reliability diagram



Calibration: reliability diagram

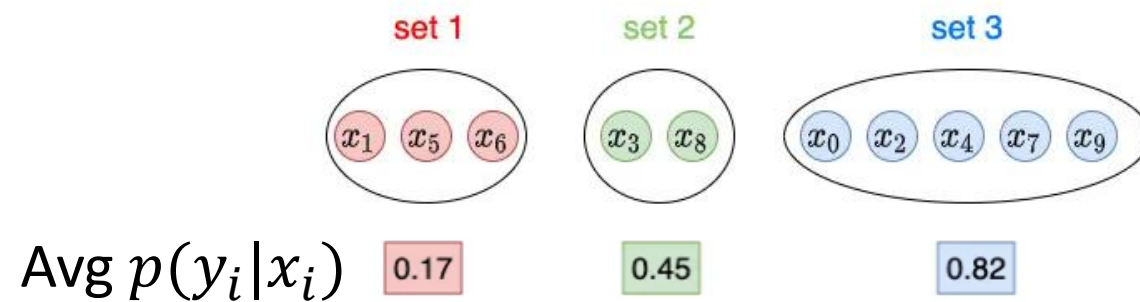
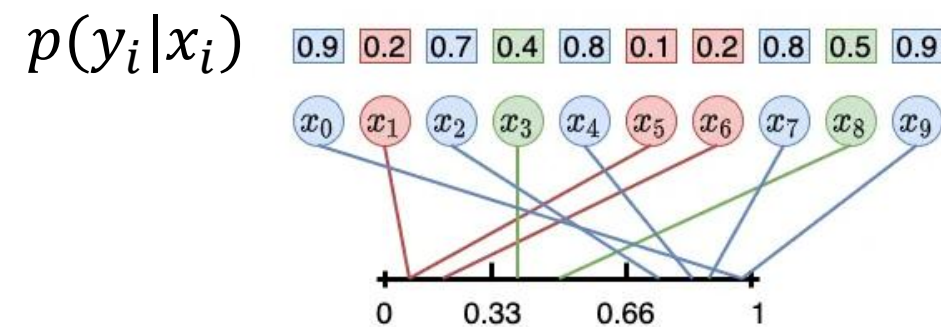
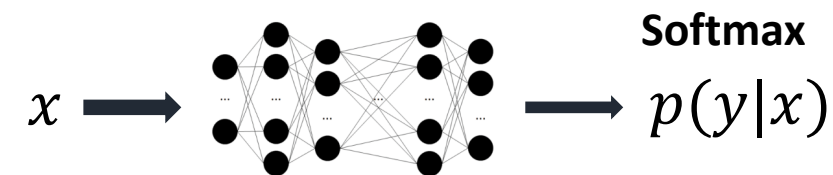
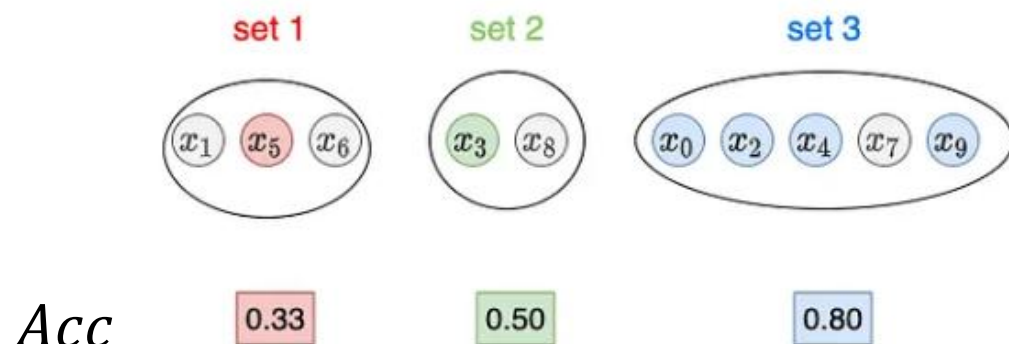


Calibration: reliability diagram

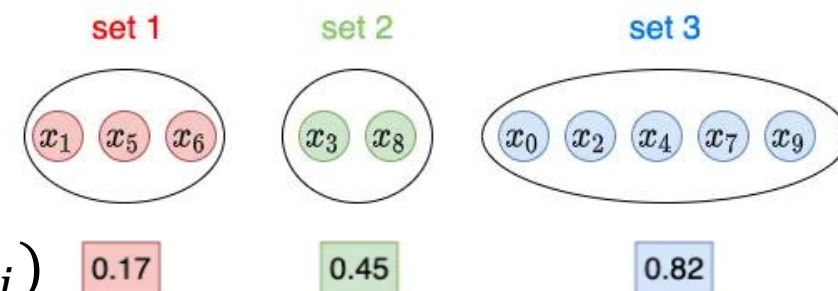
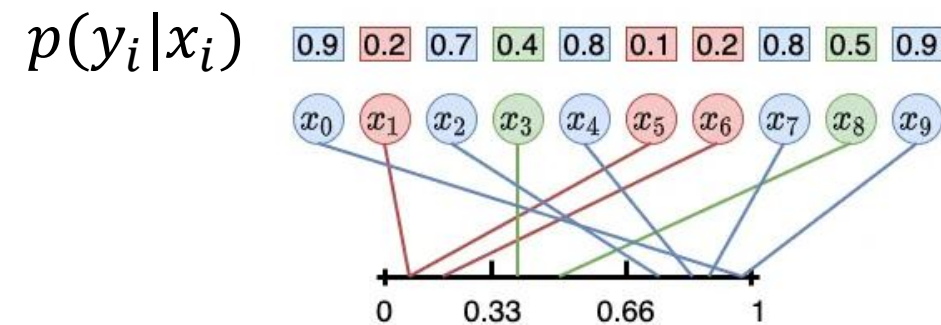
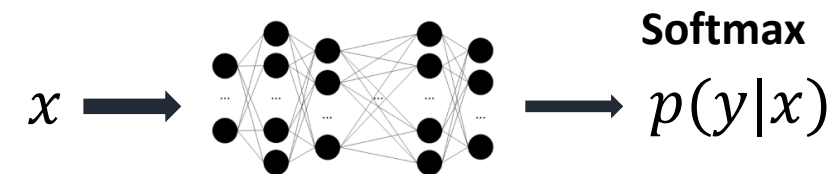
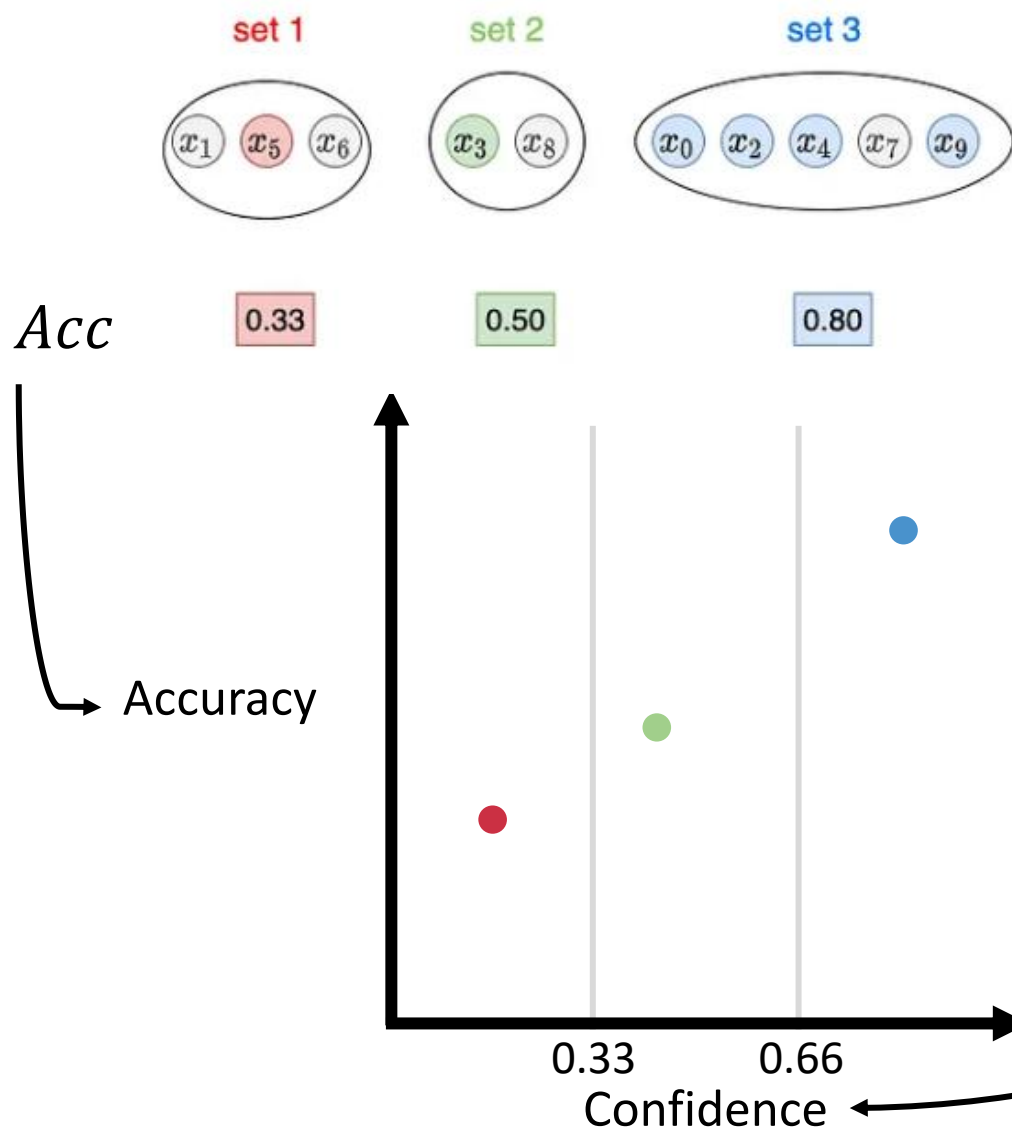


Avg $p(y_i|x_i)$ 0.17 0.45 0.82

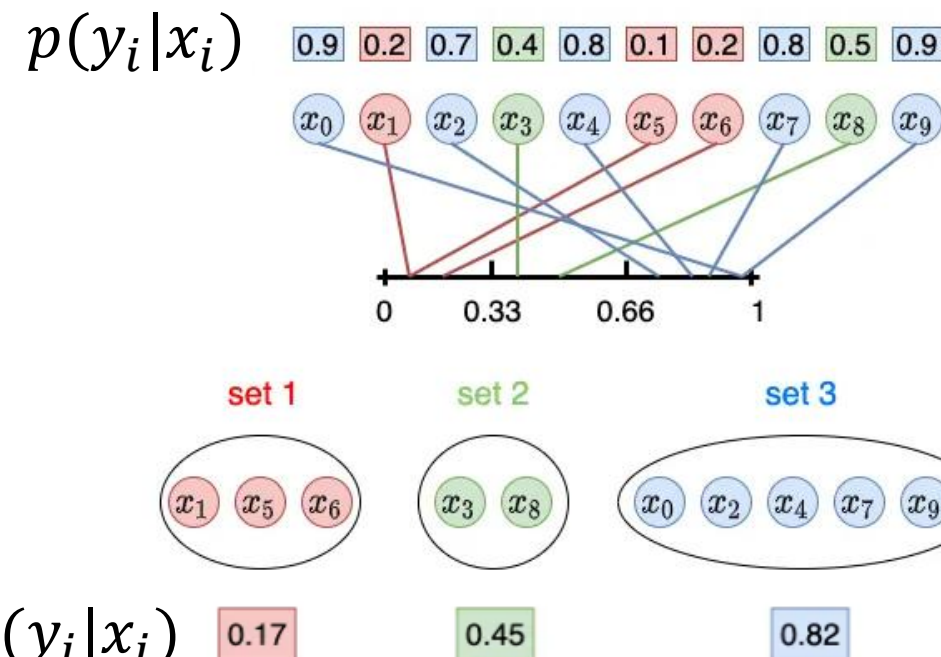
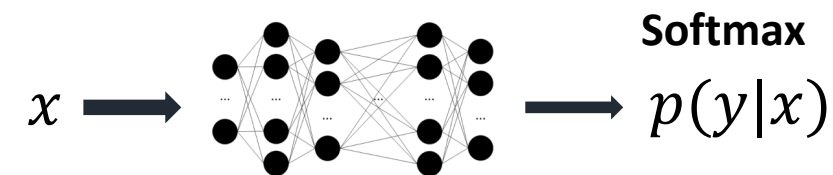
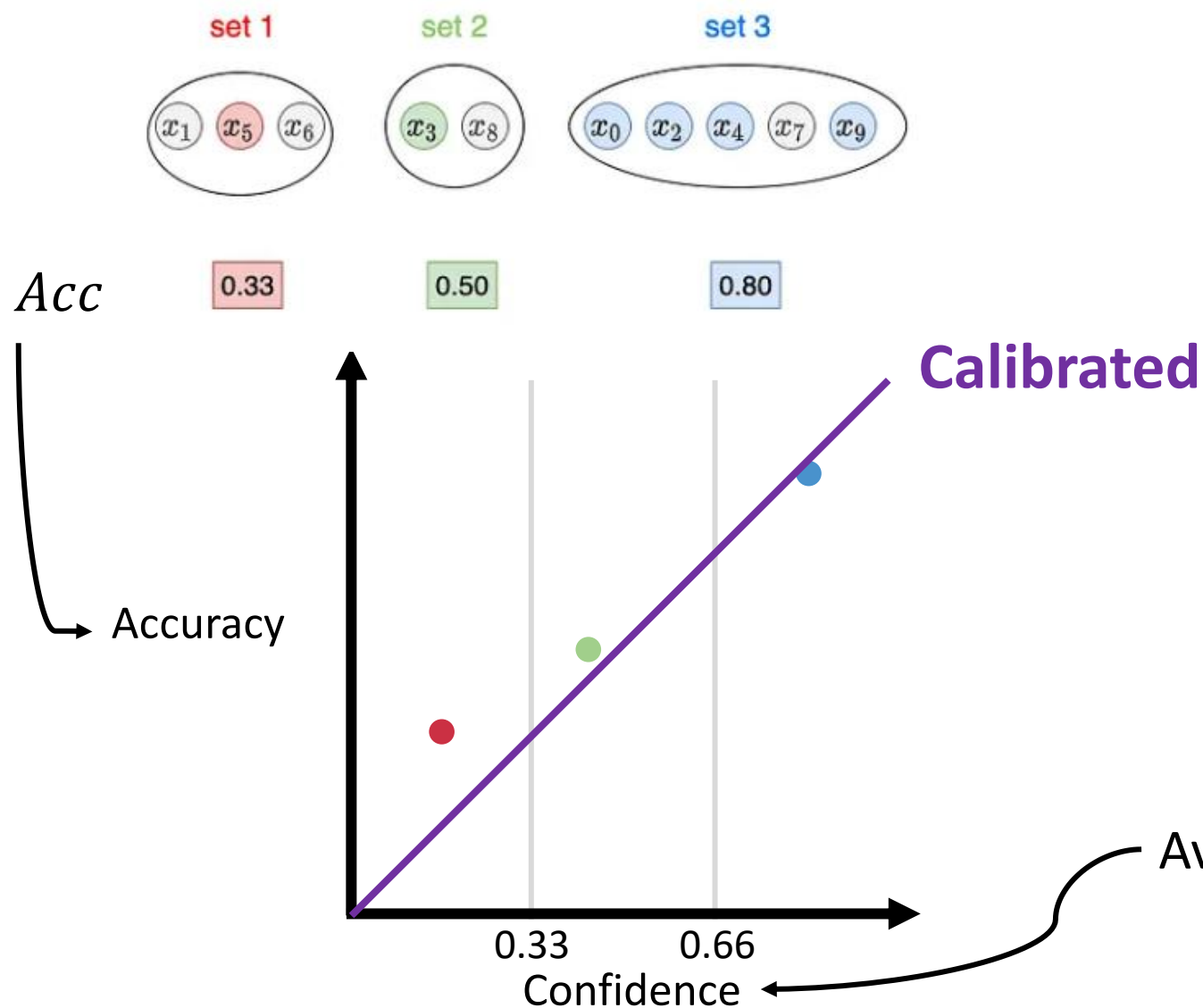
Calibration: reliability diagram



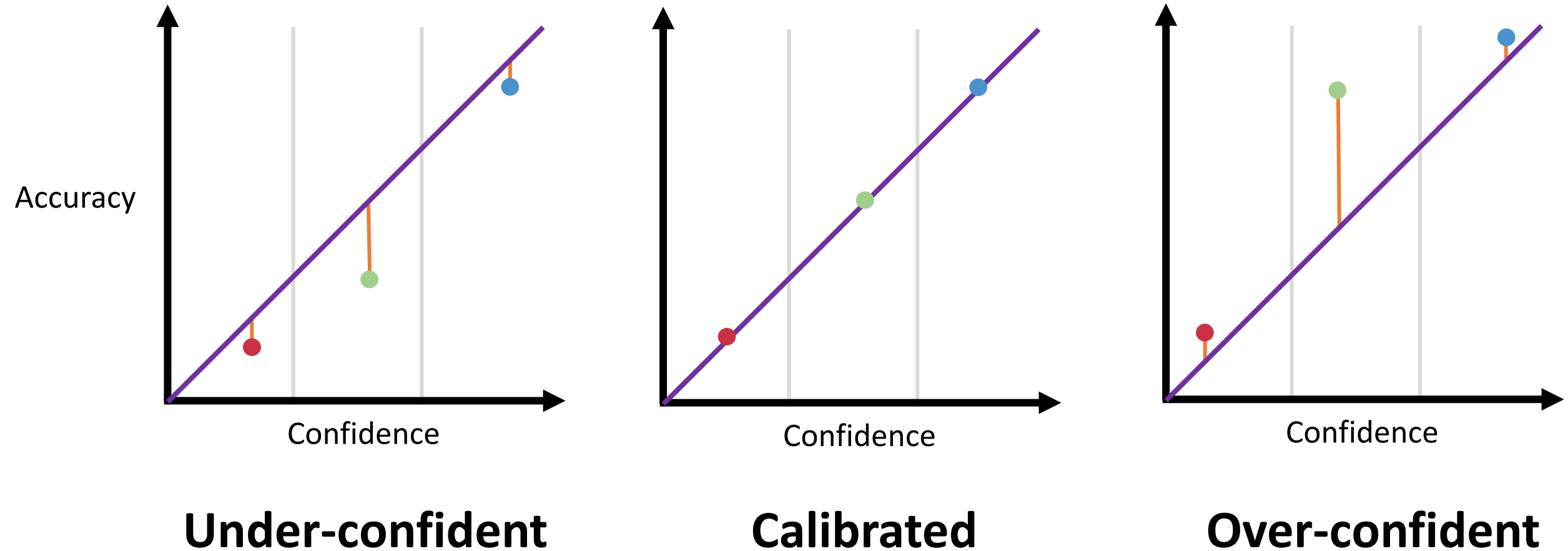
Calibration: reliability diagram



Calibration: reliability diagram

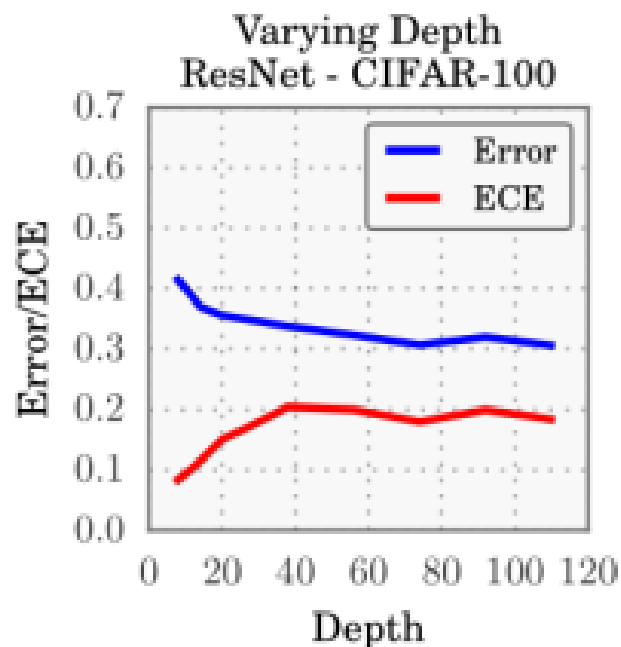


Calibration: reliability diagram

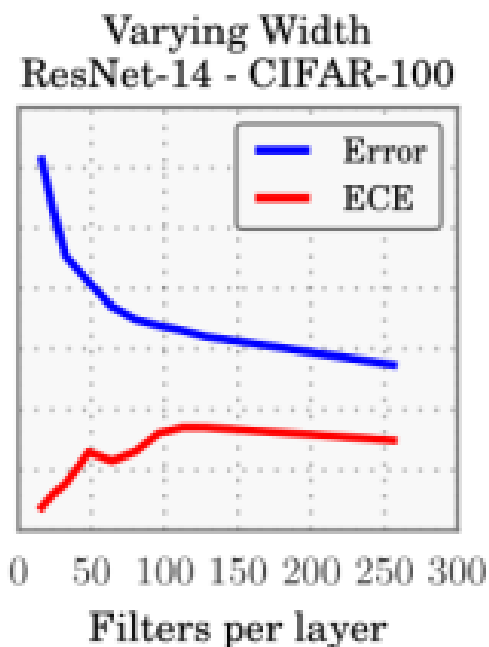


Expected Calibration Error (ECE) : $\frac{1}{|B|} \sum_{b \in B} |conf_b - acc_b|$

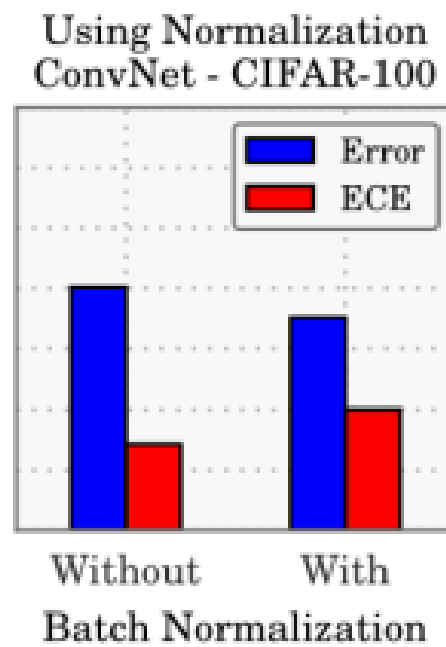
Calibration: reliability diagram



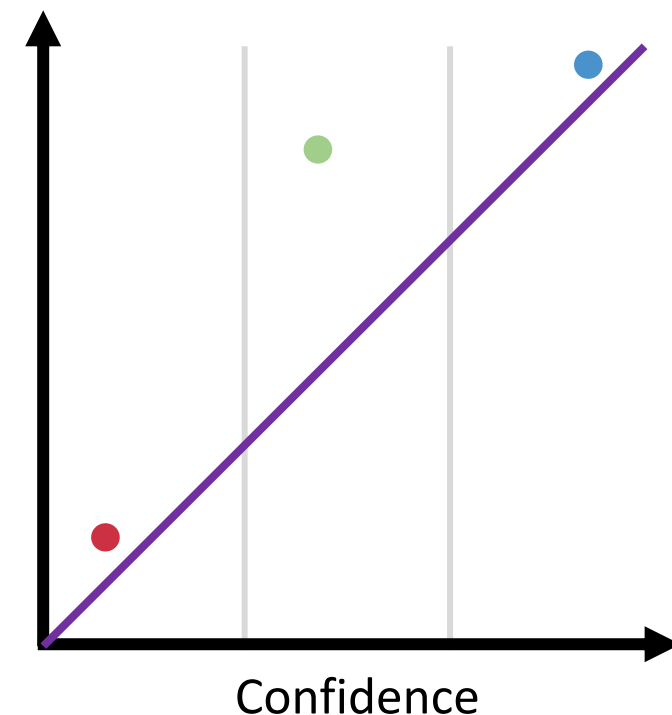
Under-confident



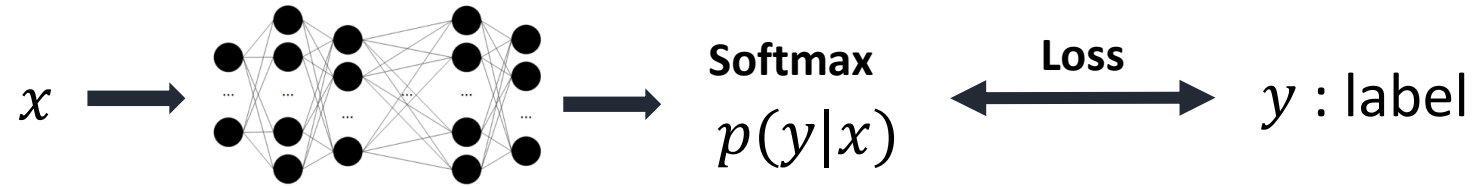
Calibrated



Over-confident



How to calibrate?



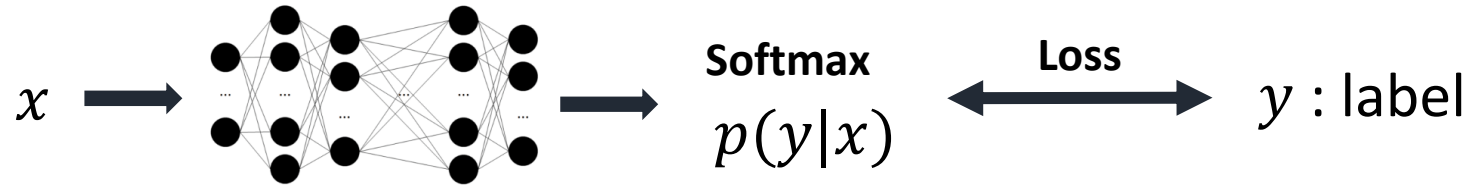
Label smoothing

Cross-entropy loss:

$$CE = - \sum_{i=1}^K y_i \log(p_i)$$

$$\text{with } y_i = \frac{\varepsilon/K}{1-\varepsilon}$$

How to calibrate?



Label smoothing

Cross-entropy loss:

$$CE = - \sum_{i=1}^K y_i \log(p_i)$$

with $y_i = \frac{\varepsilon/K}{1-\varepsilon}$

Temperature scaling

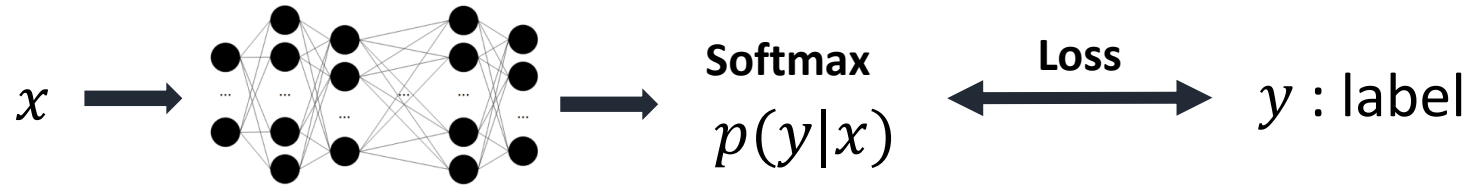
Softmax:

$$\frac{e^{z_k}}{\sum_i e^{z_i}} \rightarrow \frac{e^{z_k/T}}{\sum_i e^{z_i/T}}$$

T chosen on the validation set:

$$\min_T NLL(T)$$

How to calibrate?



Label smoothing

Cross-entropy loss:

$$CE = - \sum_{i=1}^K y_i \log(p_i)$$

with $y_i = \frac{\varepsilon/K}{1-\varepsilon}$

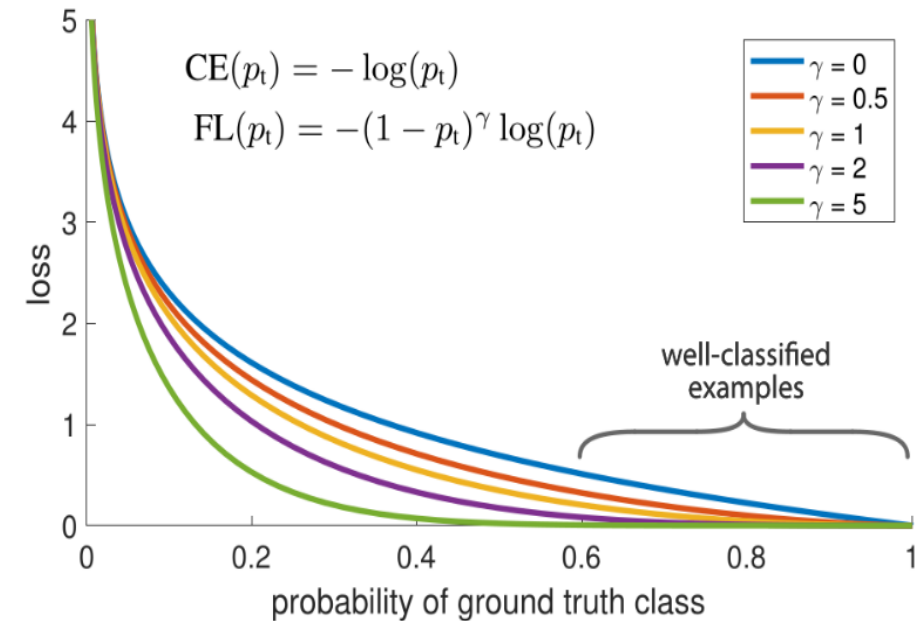
Temperature scaling

Softmax:

$$\frac{e^{z_k}}{\sum_i e^{z_i}} \rightarrow \frac{e^{z_k/T}}{\sum_i e^{z_i/T}}$$

T chosen on the validation set:
 $\min_T NLL(T)$

Focal loss

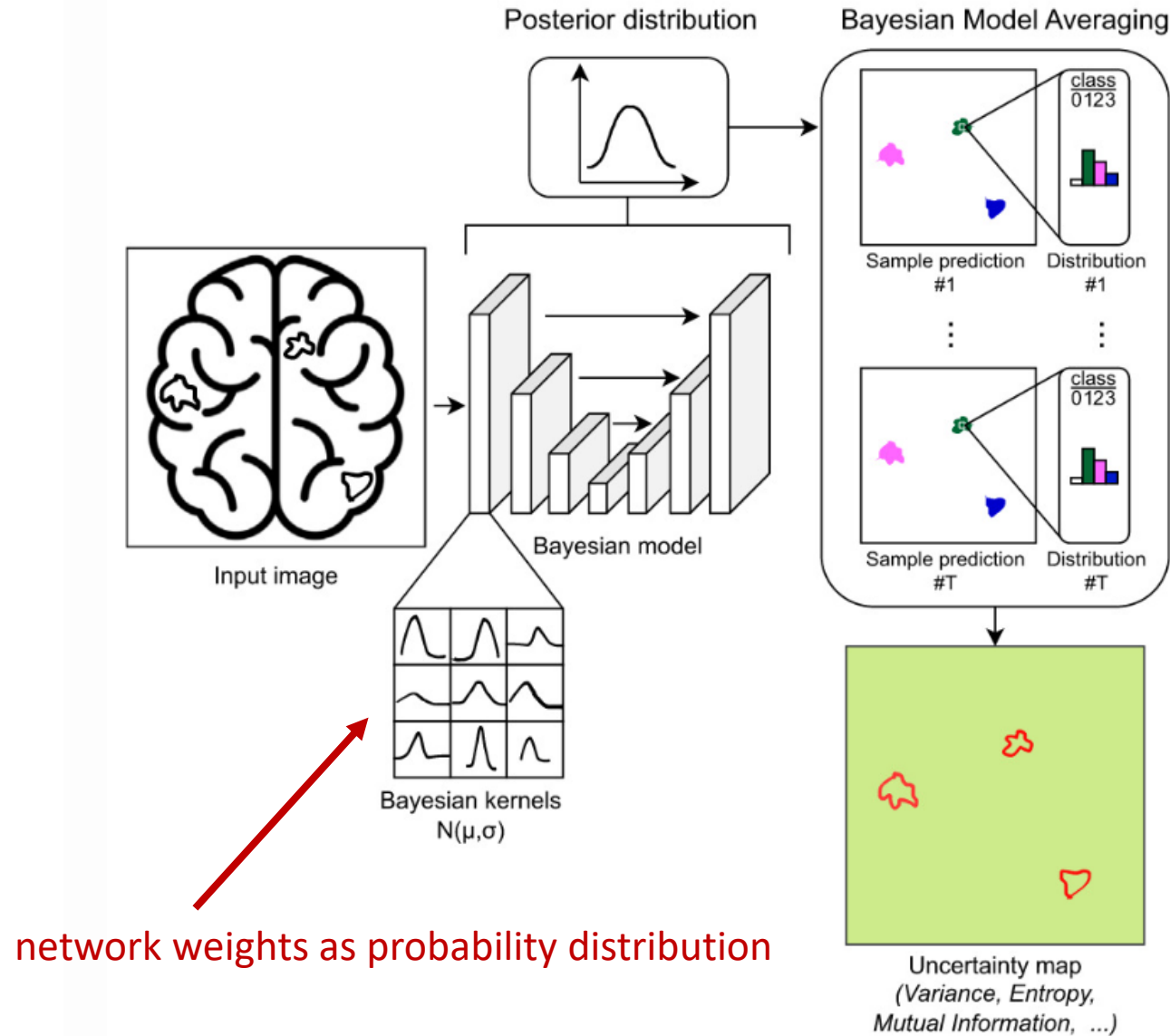


Szegedy et al., CVPR, 2016

Guo et al., ICML, 2017

Mukhoti et al., Neurips, 2020

Quantification: bayesian neural networks



Advantages

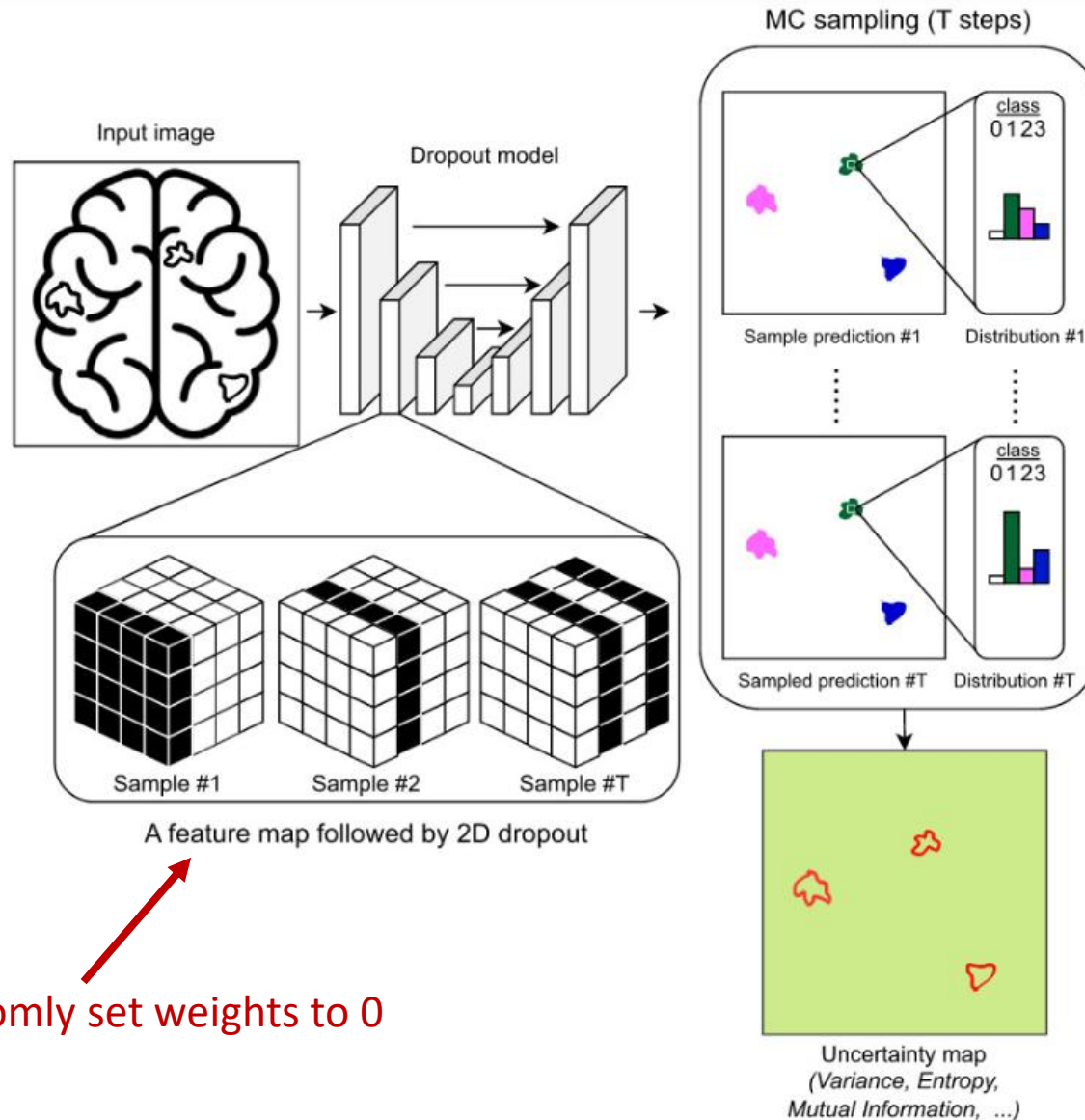
- Principled probabilistic framework
- Strong theoretical grounding

Limitations

- Depends on the choice of the prior distribution
- Computationally expensive
- Fixed architecture

Blundell et al., ICML, 2015
Lambert et al., Artificial Intelligence for Medicine, 2024

Quantification: Monte-Carlo dropout



Randomly set weights to 0

Advantages

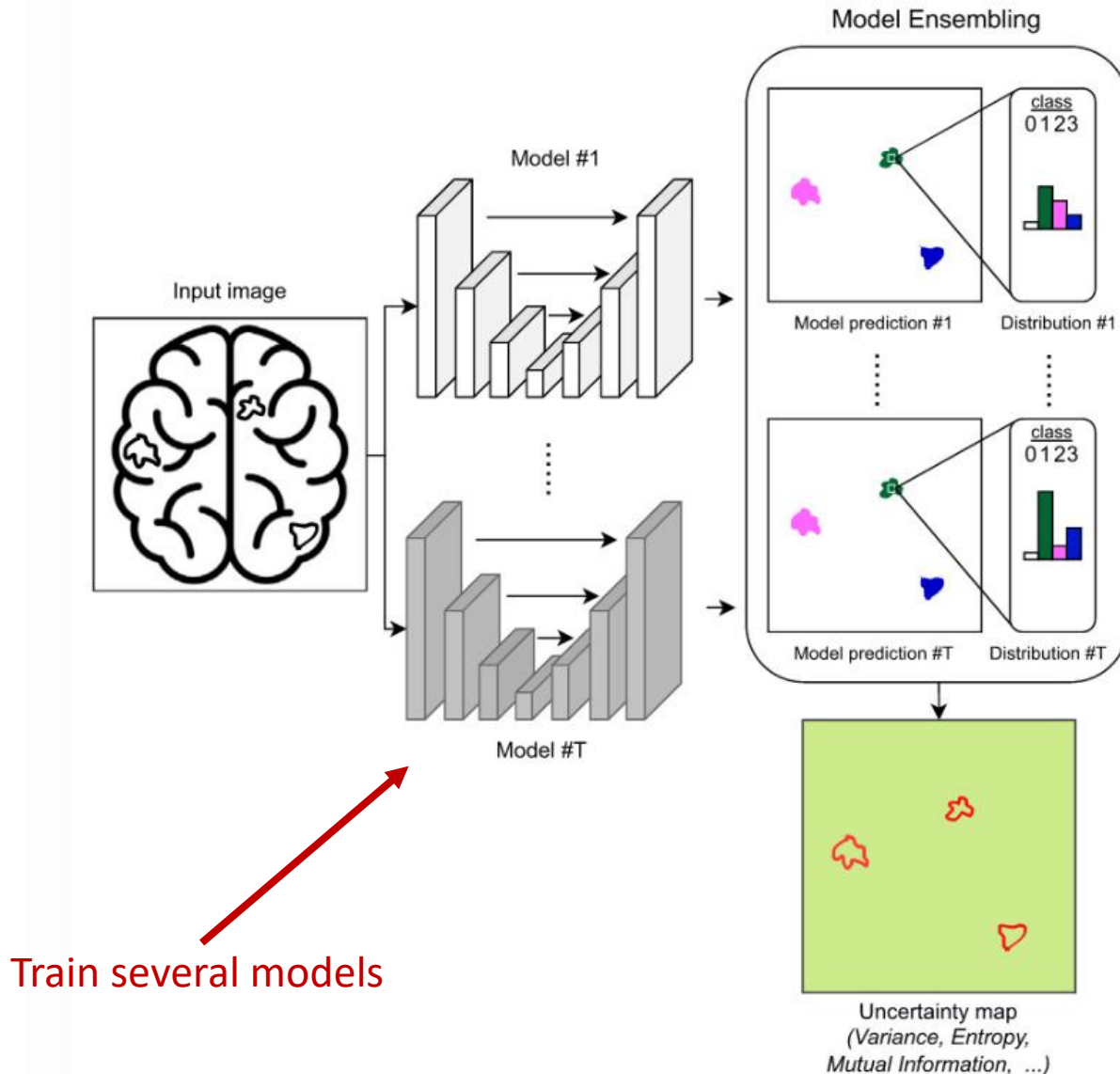
- Easy to implement
- Low overhead compared to BNN
- Works with existing trained models

Limitations

- Depends on dropout design
- Less theoretically precise than BNN

Gal & Ghahramani, ICML, 2016
Lambert et al., Artificial Intelligence for Medicine, 2024

Quantification: ensemble learning



Advantages

- Strong empirical performance
- Simple and robust
- Can boost performances

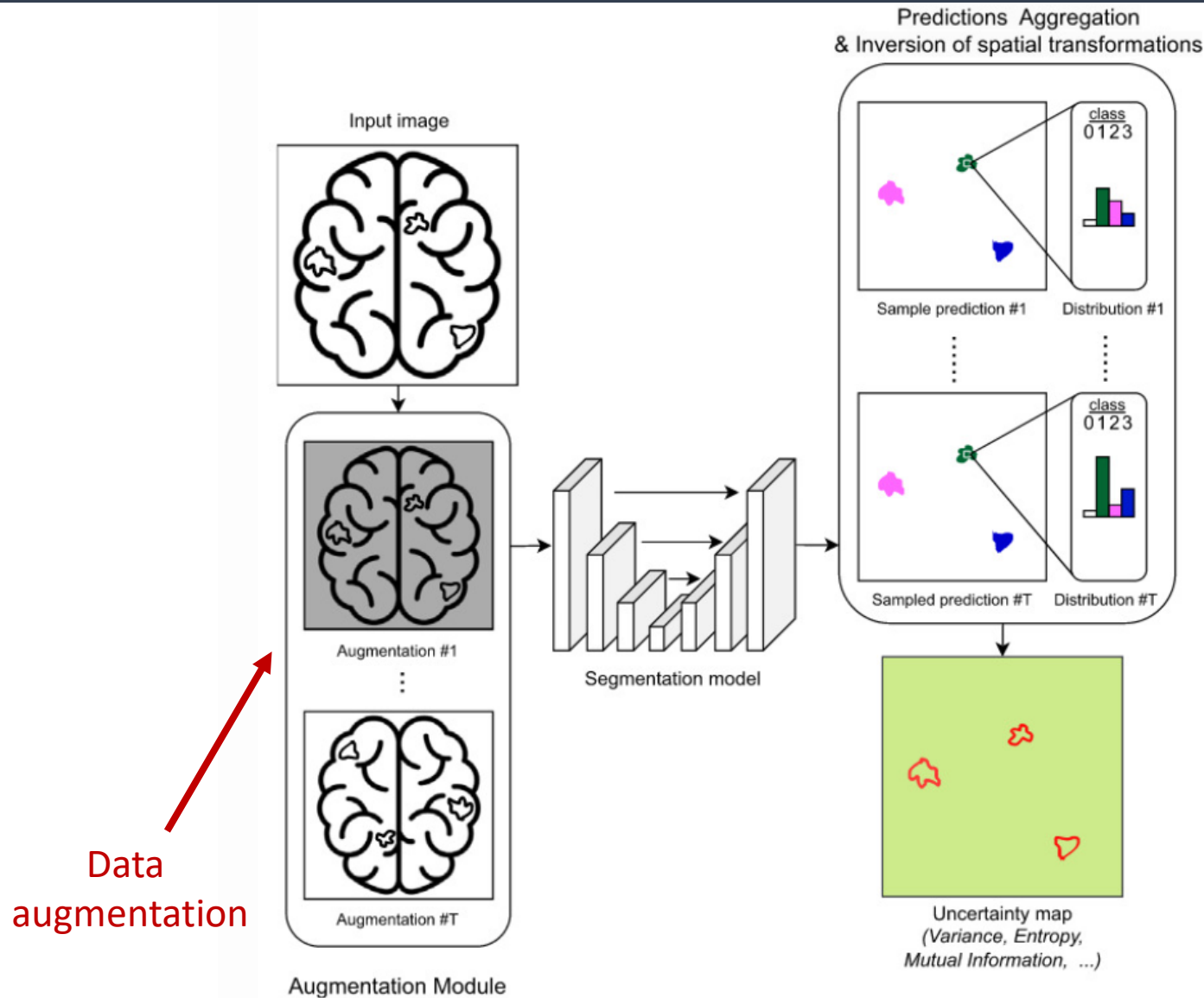
Limitations

- Depends on the models variability
- Computationally expensive

Lakshminarayanan et al., NIPS, 2017

Lambert et al., Artificial Intelligence for Medicine, 2024

Quantification: test-time augmentation



Advantages

- Easy to apply without retraining
- Improves robustness and accuracy

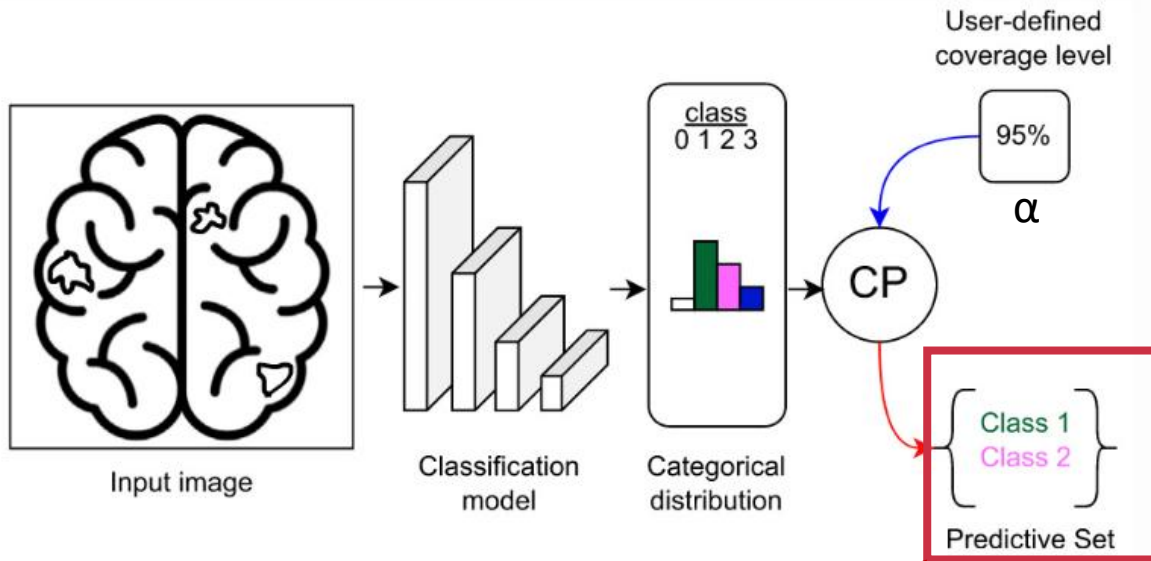
Limitations

- Depends on the transformations
- Does not capture model uncertainty

Ayhan et Berens. MIDL, 2018

Lambert et al., Artificial Intelligence for Medicine, 2024

Quantification: conformal prediction



For a test image I , find a set of labels $C(I)$ such that: $P(y \in C(I)) \geq 1 - \alpha$



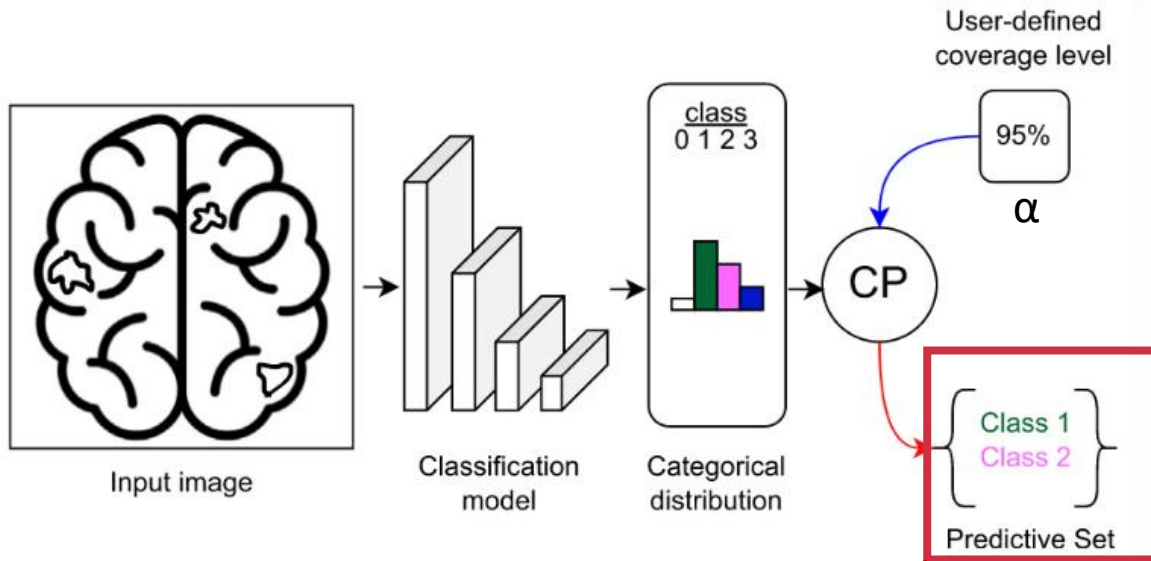
{ fox
squirrel
0.99 }



{ fox
squirrel, gray
0.82, fox,
0.03 bucket,
0.02 rain
0.02 barrel }

Angelopoulos & Bates. MIDL, 2018
Lambert et al., Artificial Intelligence for Medicine, 2024

Quantification: conformal prediction



For a test image I , find a set of labels $C(I)$ such that: $P(y \in C(I)) \geq 1 - \alpha$



{ fox
squirrel
0.99 }



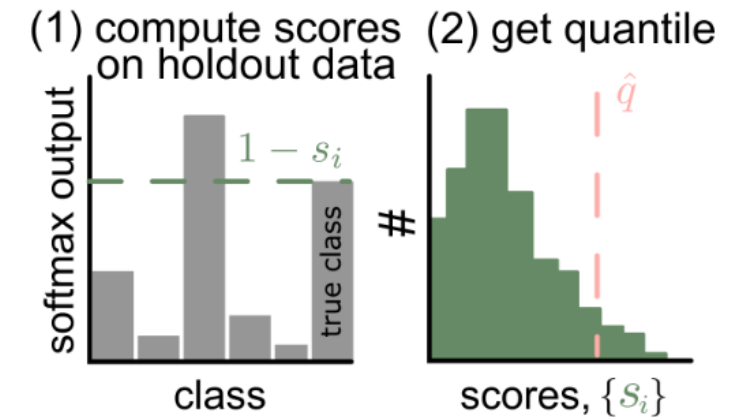
{ fox
squirrel, 0.82, gray
fox, 0.03, bucket,
rain barrel, 0.02 }

On a calibration set (size n):

(1) Compute calibration scores for each class i :

$$s_i = 1 - F_{y_i}(x_i)$$

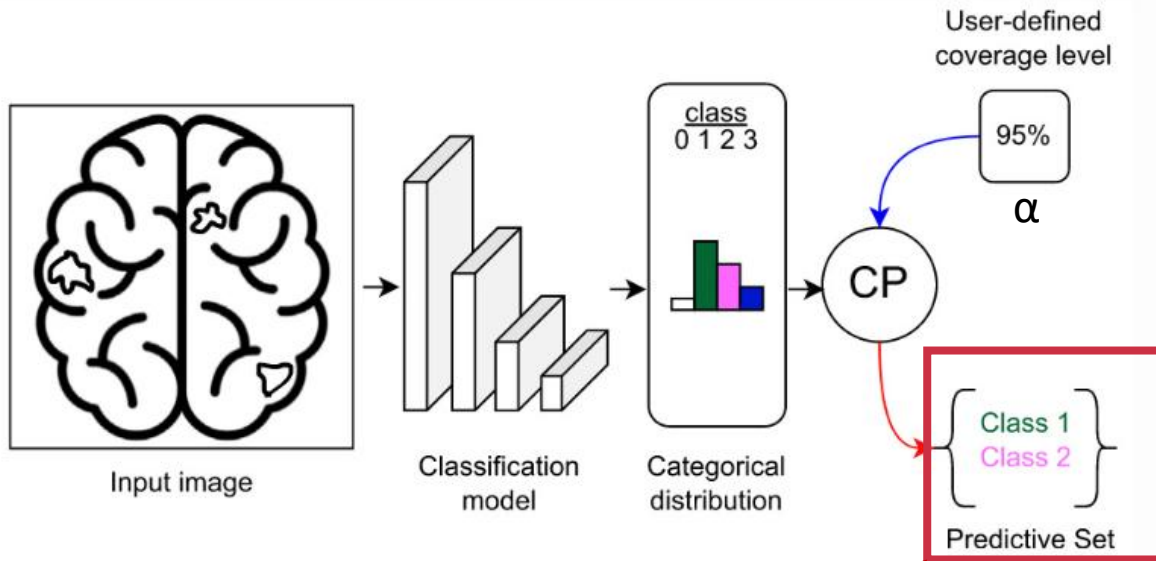
(2) Compute $\hat{q} = \lceil (n + 1)(1 - \alpha) \rceil / n$ quantile of $\{s_i\}$



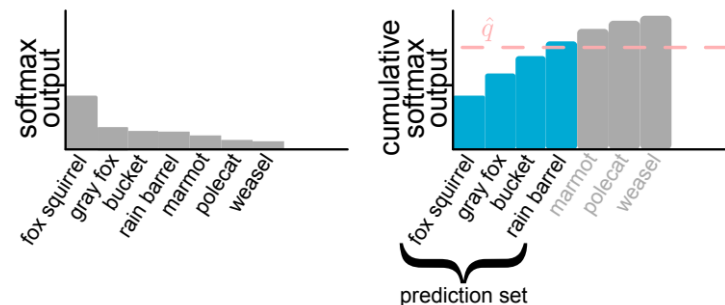
Angelopoulos & Bates. MIDL, 2018

Lambert et al., Artificial Intelligence for Medicine, 2024

Quantification: conformal prediction



For a test image I , find a set of labels $C(I)$ such that: $P(y \in C(I)) \geq 1 - \alpha$

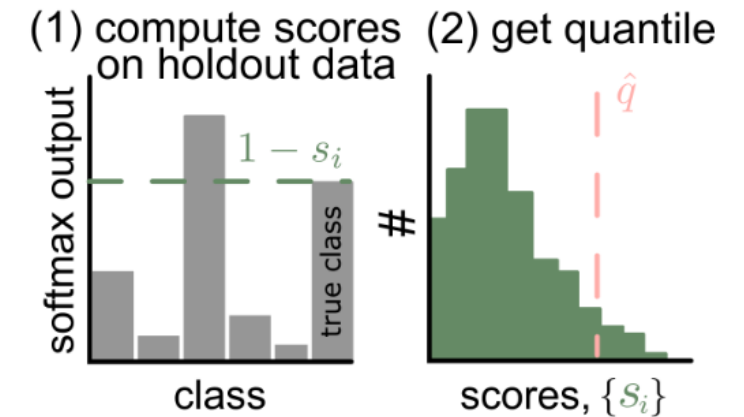


On a calibration set (size n):

(1) Compute calibration scores for each class i :

$$s_i = 1 - F_{y_i}(x_i)$$

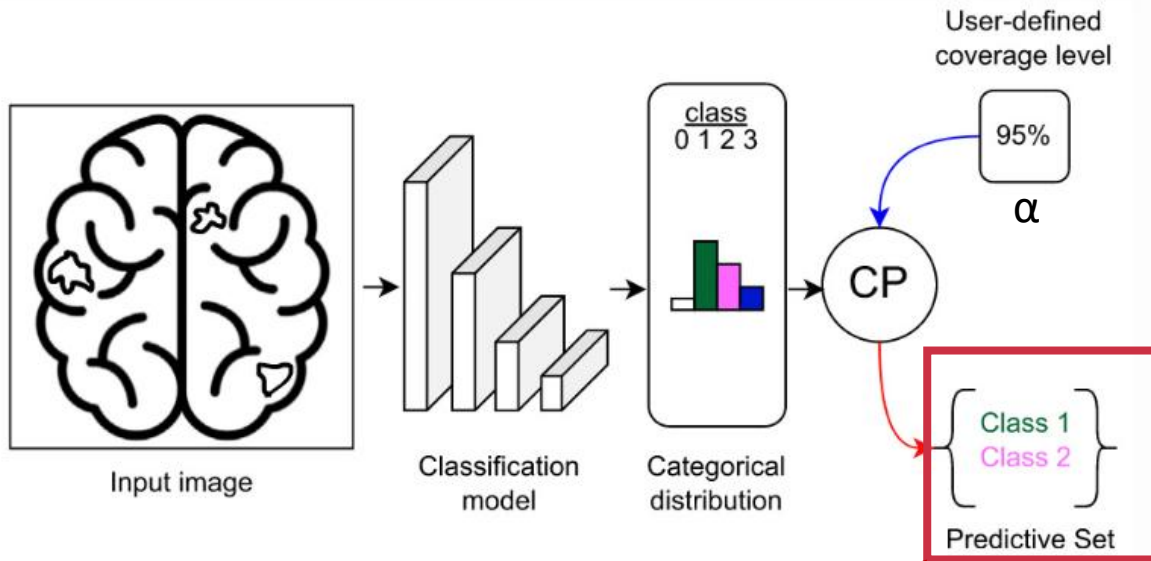
(2) Compute $\hat{q} = \lceil (n + 1)(1 - \alpha) \rceil / n$ quantile of $\{s_i\}$



Test set

Angelopoulos & Bates. MIDL, 2018
Lambert et al., Artificial Intelligence for Medicine, 2024

Quantification: conformal prediction



For a test image I , find a set of labels $C(I)$ such that: $P(y \in C(I)) \geq 1 - \alpha$



{ fox
squirrel
0.99 }



{ fox
squirrel, gray
fox, bucket, rain
0.82, 0.03, 0.02, barrel
0.02 }

Advantages

- Distribution-free coverage guarantees
- Can be applied to any model
- Can be combined with another uncertainty quantification method

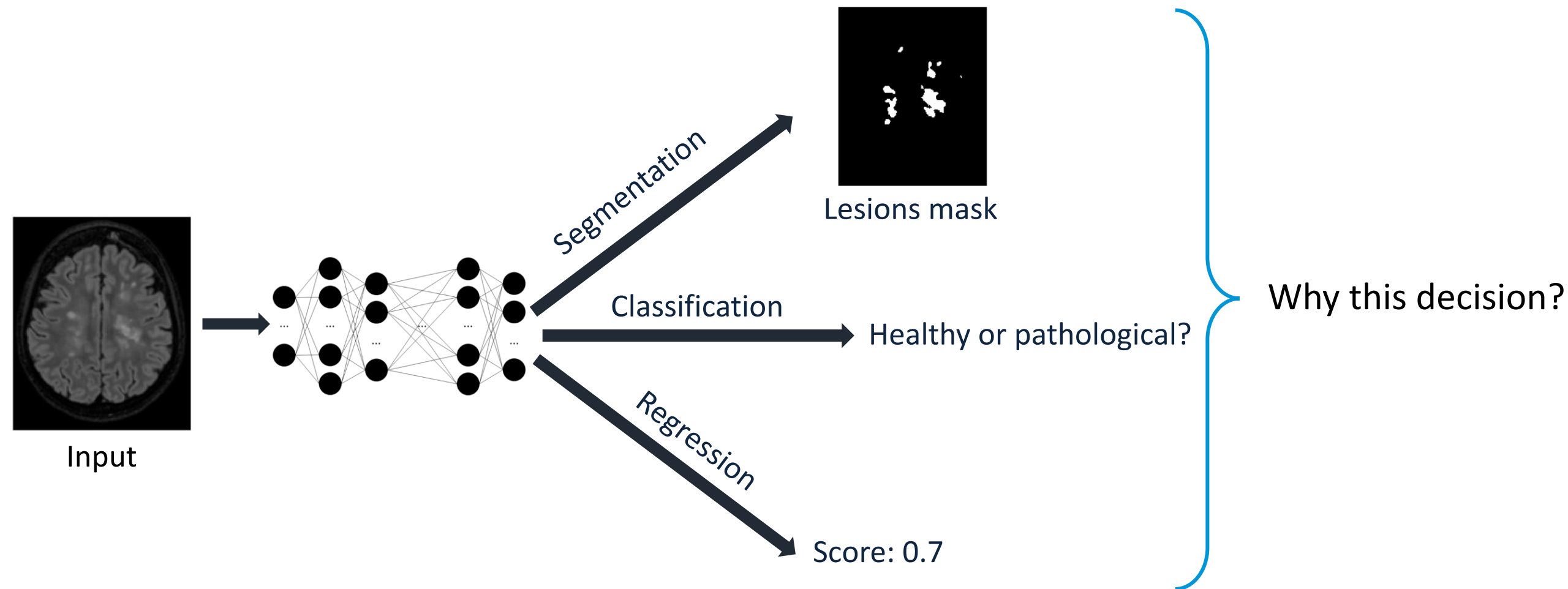
Limitations

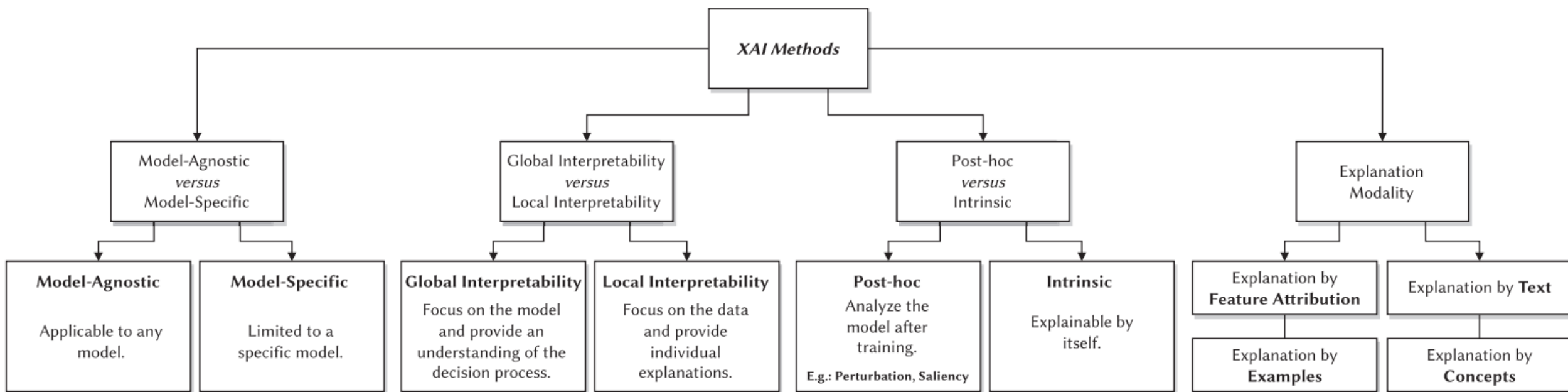
- Produces sets that could be large
- Requires separate calibration dataset

Angelopoulos & Bates. MIDL, 2018

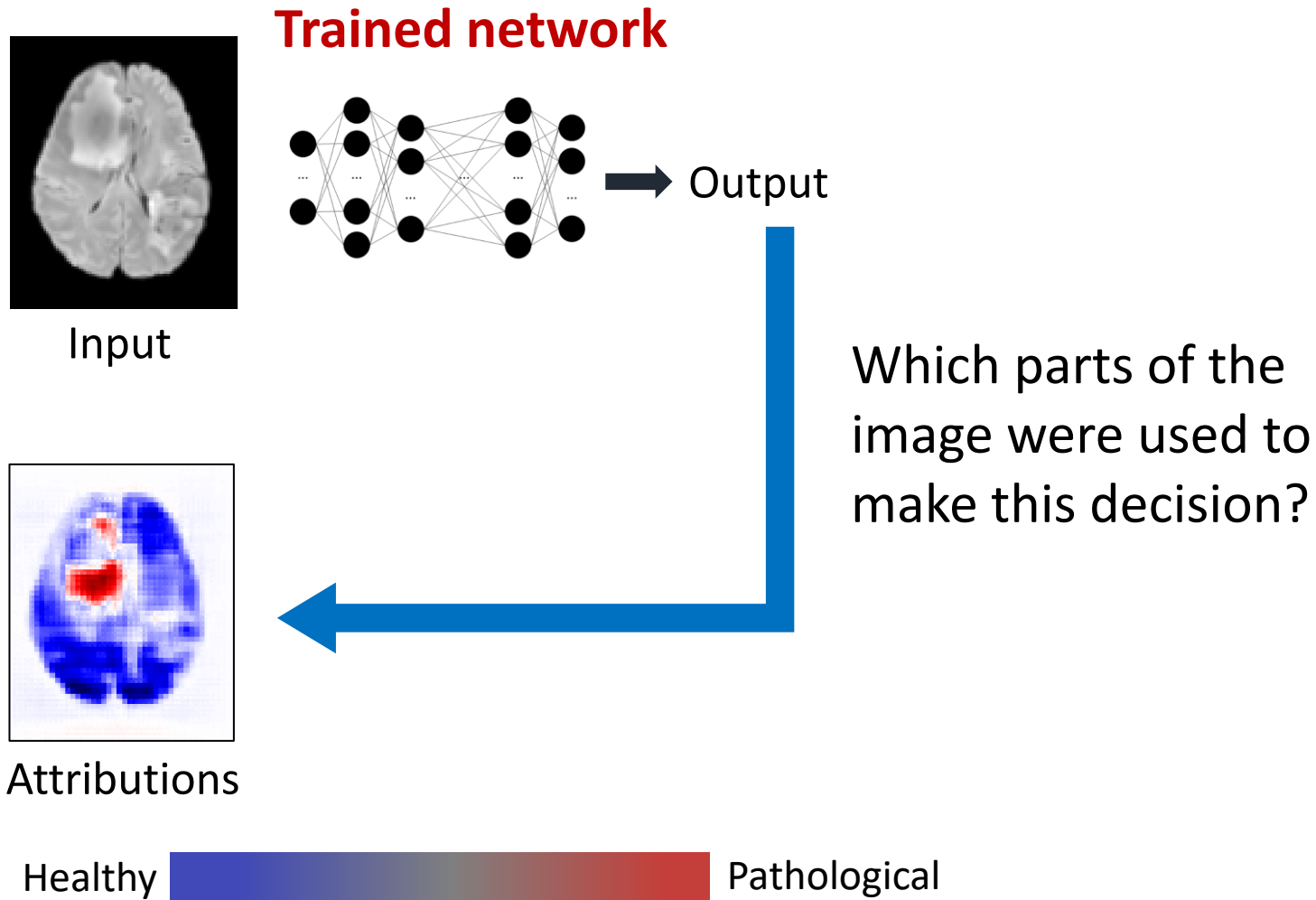
Lambert et al., Artificial Intelligence for Medicine, 2024

Explainability





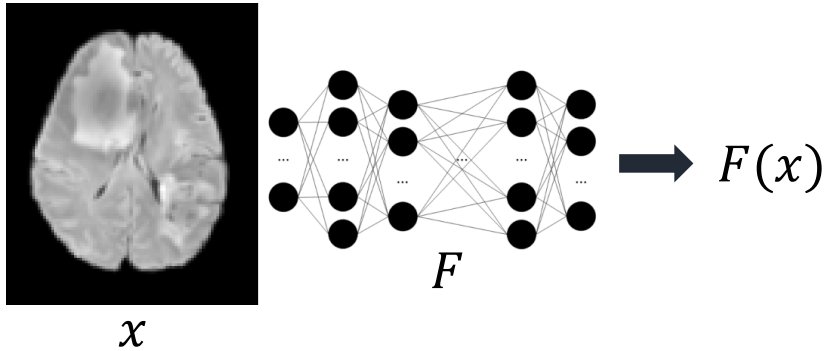
Attributions maps



Based on:

- Gradients
- Activations
- Perturbations

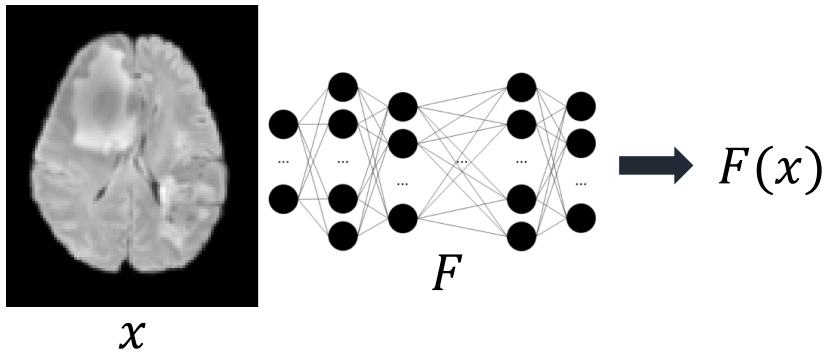
Gradient-based attributions



Gradient (G):

$$A_i^G(x) = \frac{\partial F(x)}{\partial x_i}$$

Gradient-based attributions



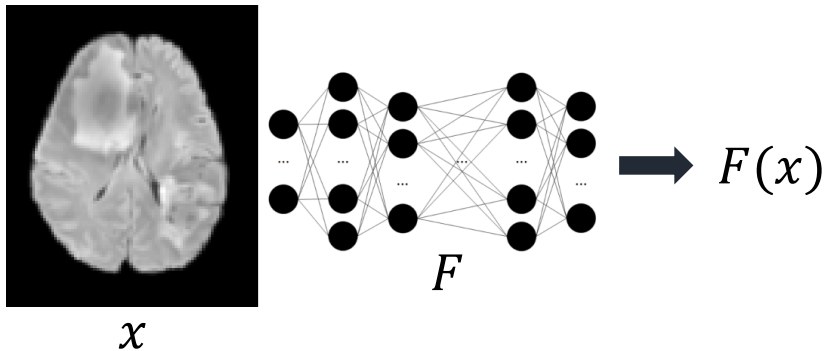
Gradient (G):

$$A_i^G(x) = \frac{\partial F(x)}{\partial x_i} \quad \text{Local}$$

Input X Gradient (IXG):

$$A_i^{I \times G}(x) = x \times A_i^G(x) \quad \text{Global}$$

Gradient-based attributions



Gradient (G):

$$A_i^G(x) = \frac{\partial F(x)}{\partial x_i} \quad \text{Local}$$

Integrated Gradients (IG)

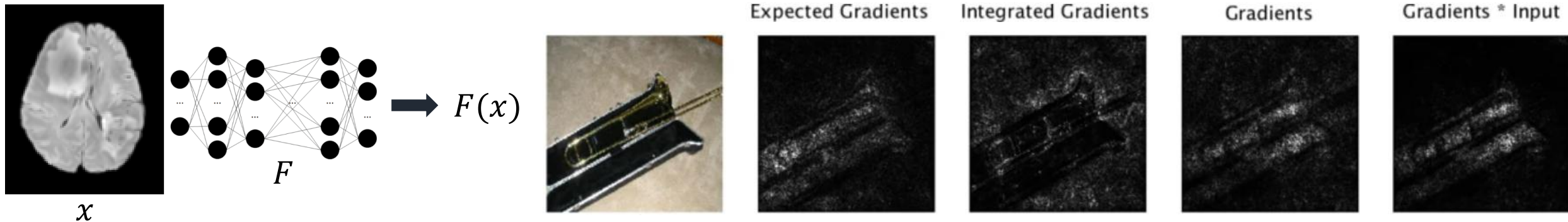
$$A_i^{IG}(x, x') = \int_{\alpha=0}^1 A_i^G(x' + \alpha(x - \underline{x'})) d\alpha \quad \text{Path}$$

Baseline

Input X Gradient (IXG):

$$A_i^{I \times G}(x) = x \times A_i^G(x) \quad \text{Global}$$

Gradient-based attributions



Gradient (G):

$$A_i^G(x) = \frac{\partial F(x)}{\partial x_i} \quad \text{Local}$$

Integrated Gradients (IG)

$$A_i^{IG}(x, x') = \int_{\alpha=0}^1 A_i^G(x' + \alpha(x - \underline{x'})) d\alpha \quad \text{Path Baseline}$$

Input X Gradient (IXG):

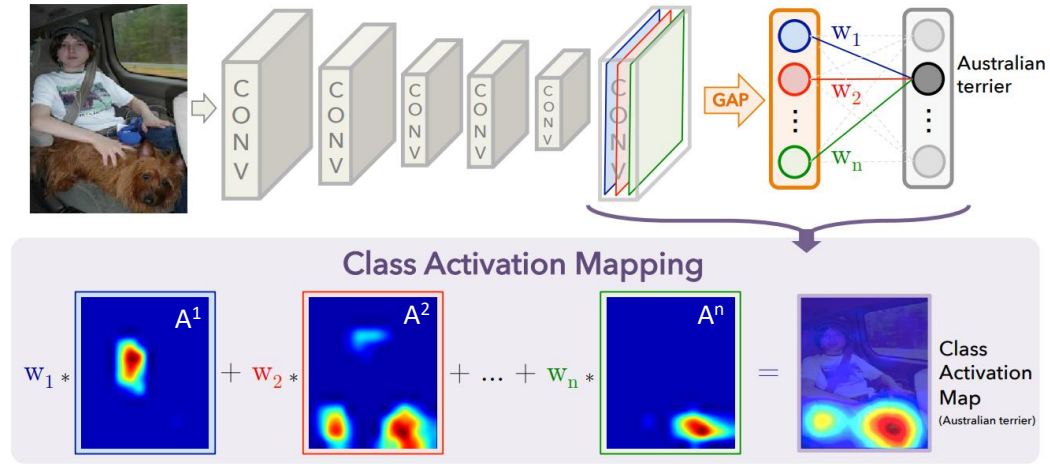
$$A_i^{I \times G}(x) = x \times A_i^G(x) \quad \text{Global}$$

Expected Gradients (EG)

$$A_i^{EG}(x) = \int_{x'} A_i^{IG}(x, x') dx' \quad \text{Baseline choice}$$

Activation-based attributions

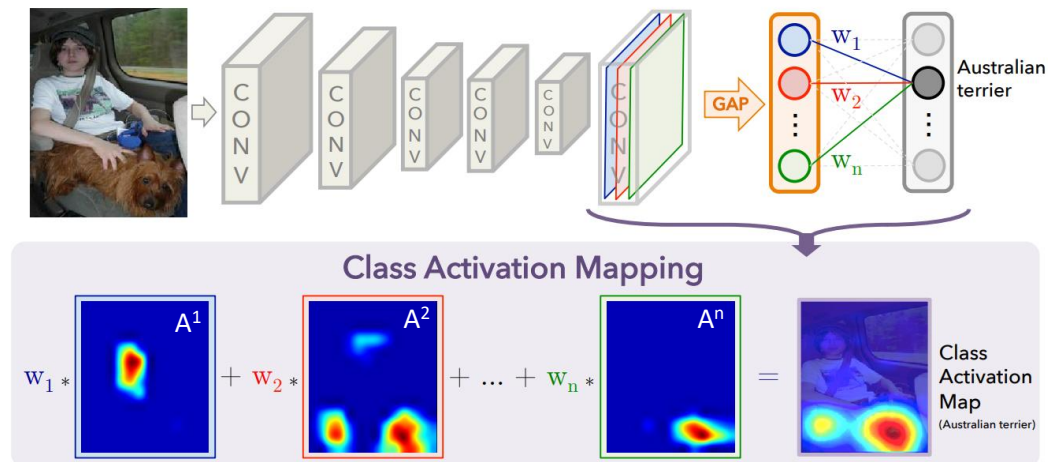
CAM:



→ specific architecture

Activation-based attributions

CAM:



→ specific architecture

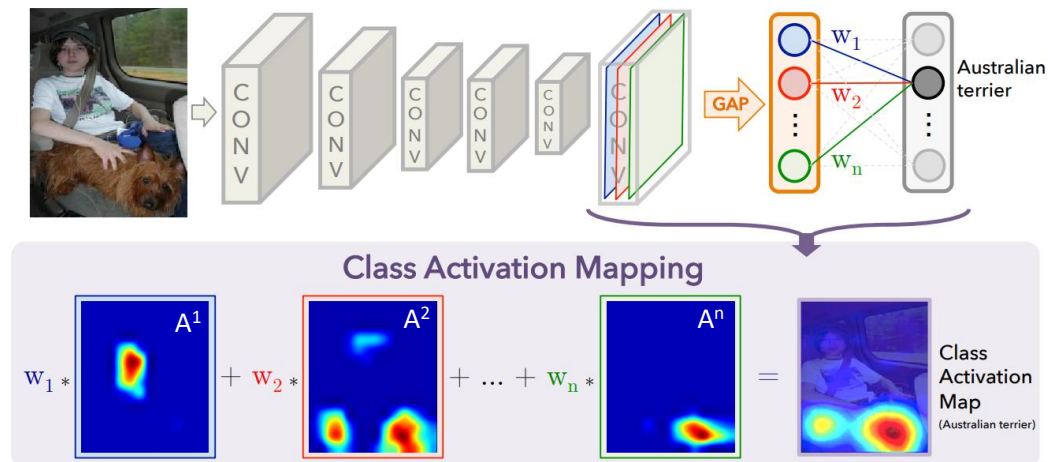
GradCAM:

$$w_k = \frac{1}{|Z|} \sum_{i \in Z} \frac{\partial F(x)}{\partial A_i^k}$$

Can be applied on every layer
→ generally last one: semantic ↗
resolution ↘

Activation-based attributions

CAM:



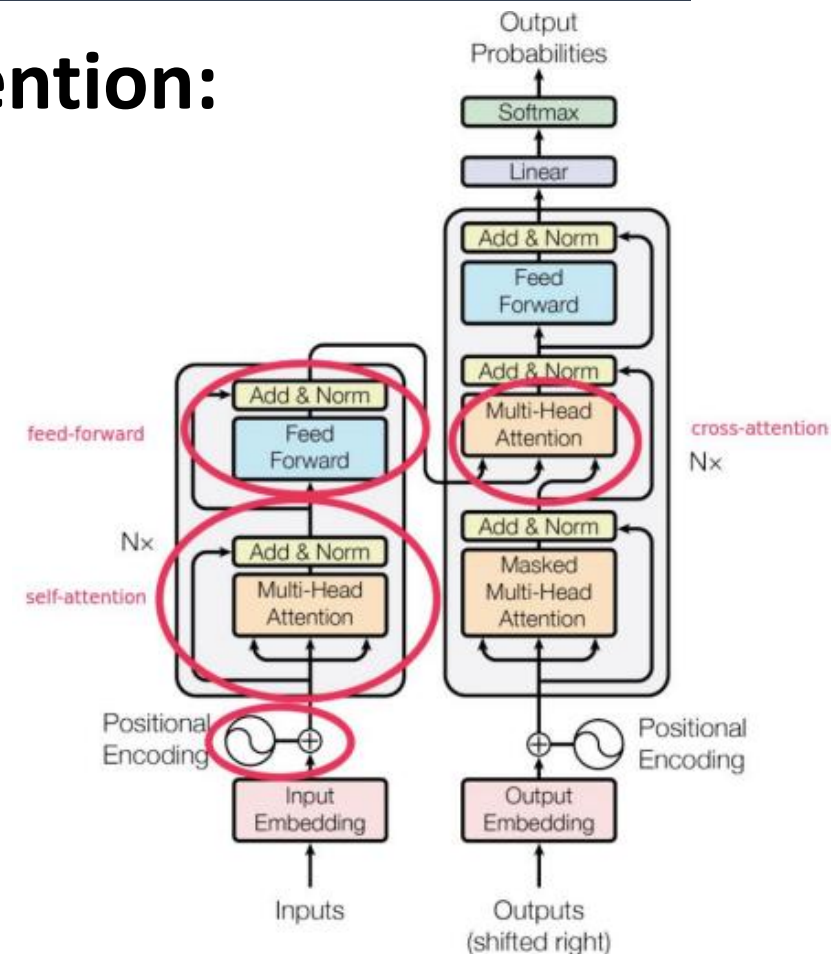
→ specific architecture

GradCAM:

$$w_k = \frac{1}{|Z|} \sum_{i \in Z} \frac{\partial F(x)}{\partial A_i^k}$$

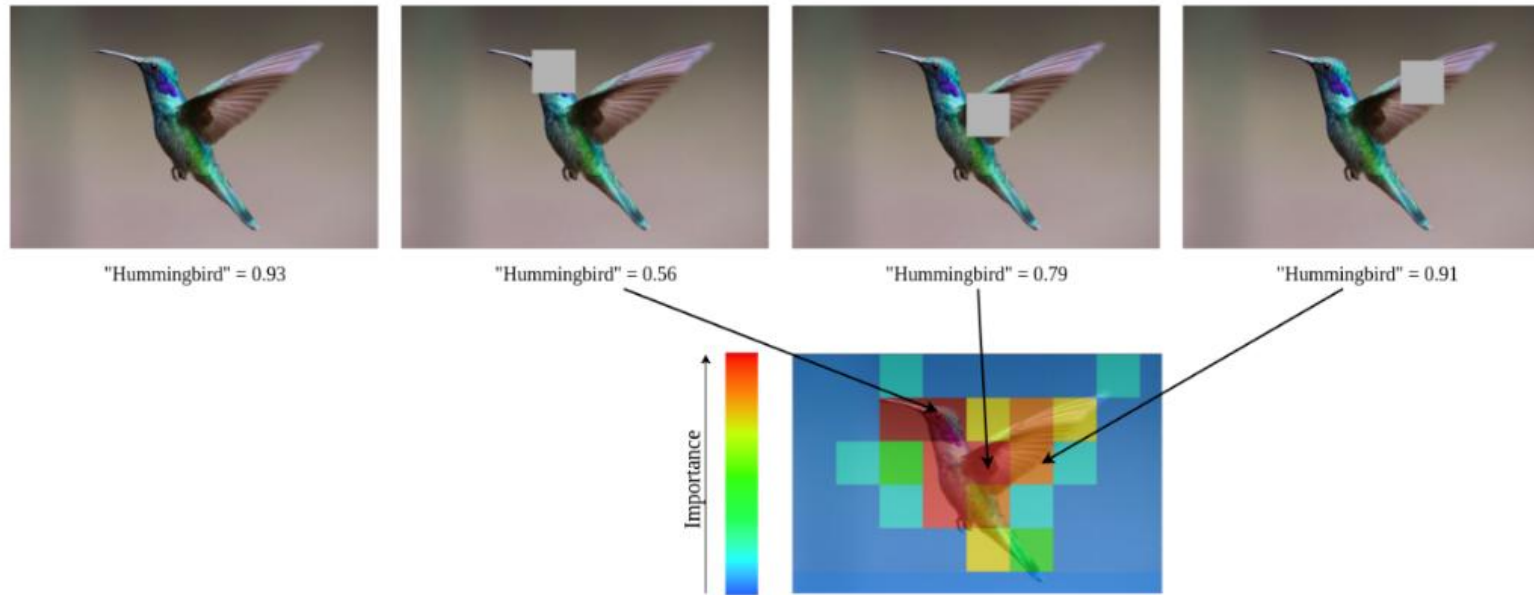
Can be applied on every layer
→ generally last one: semantic ↗
resolution ↘

Attention:



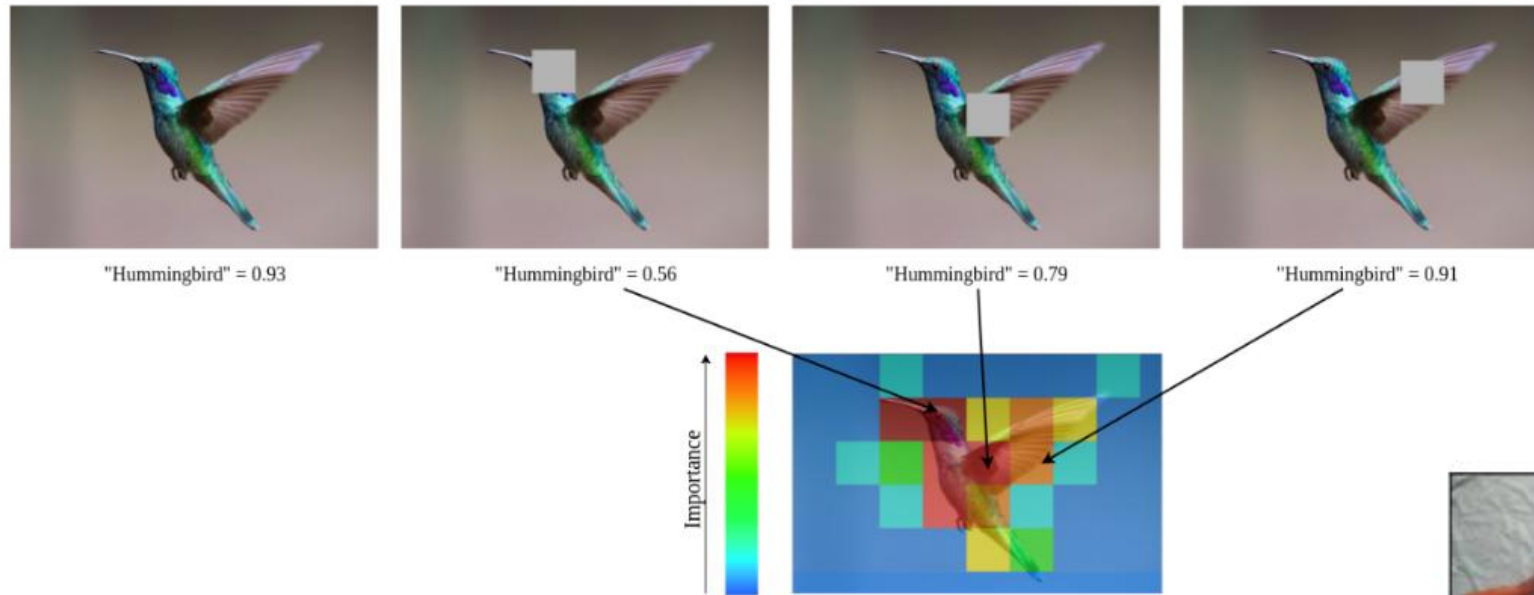
Changing attention weights $\nRightarrow$ changing predicted label
Large weights $\neq$ meaningful explanations

Perturbation-based attributions

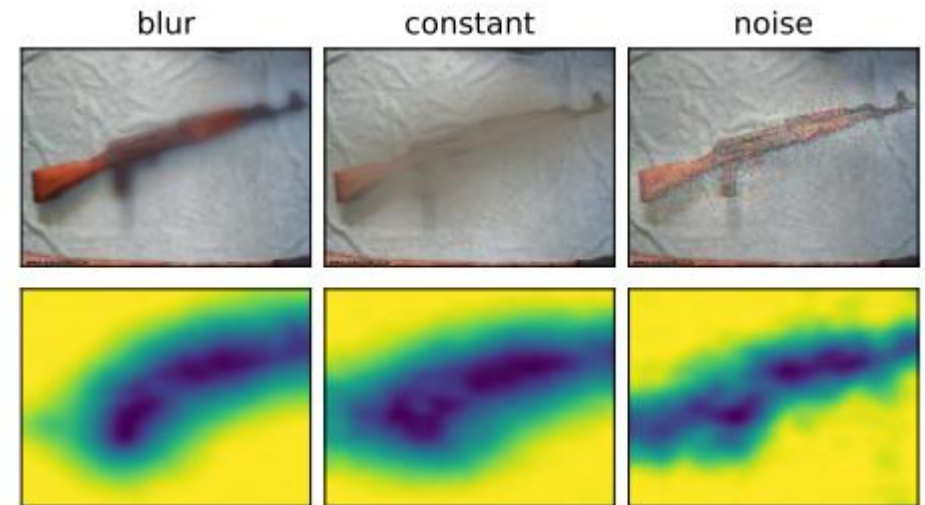


$$A_i^P(x) = F(x) - F(x^i)$$

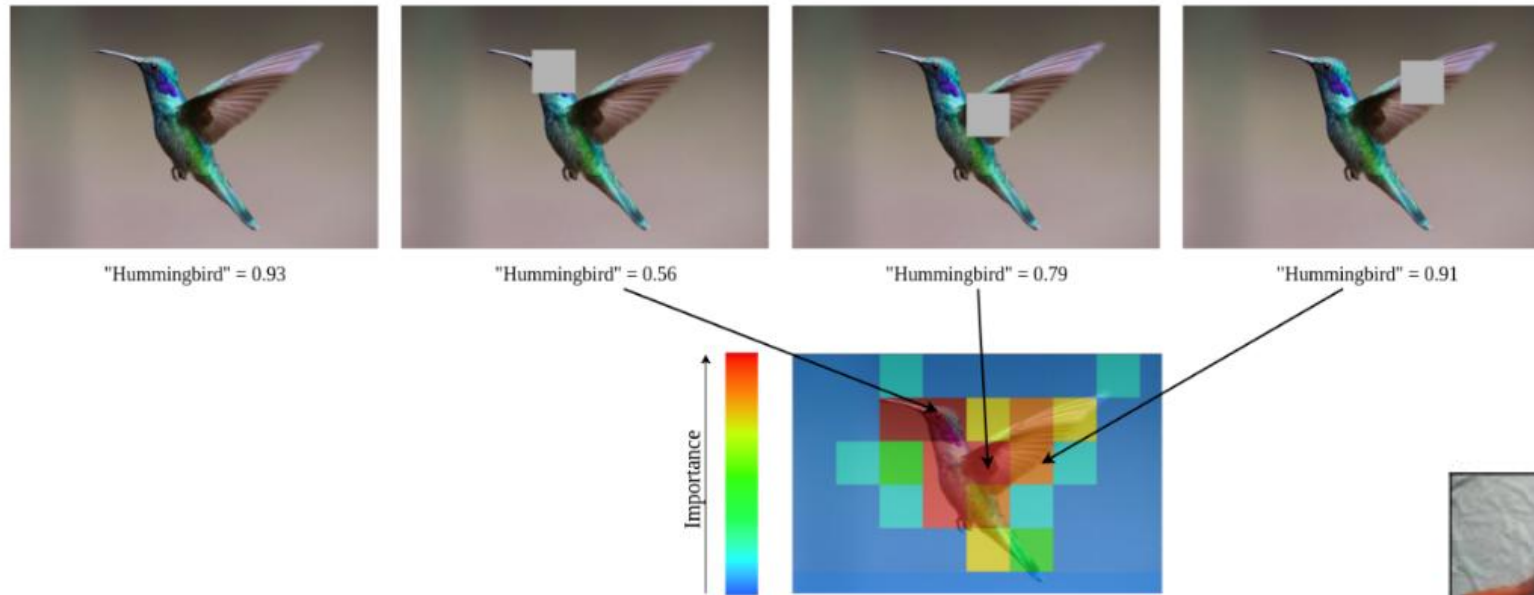
Perturbation-based attributions



$$A_i^P(x) = F(x) - F(x^i)$$



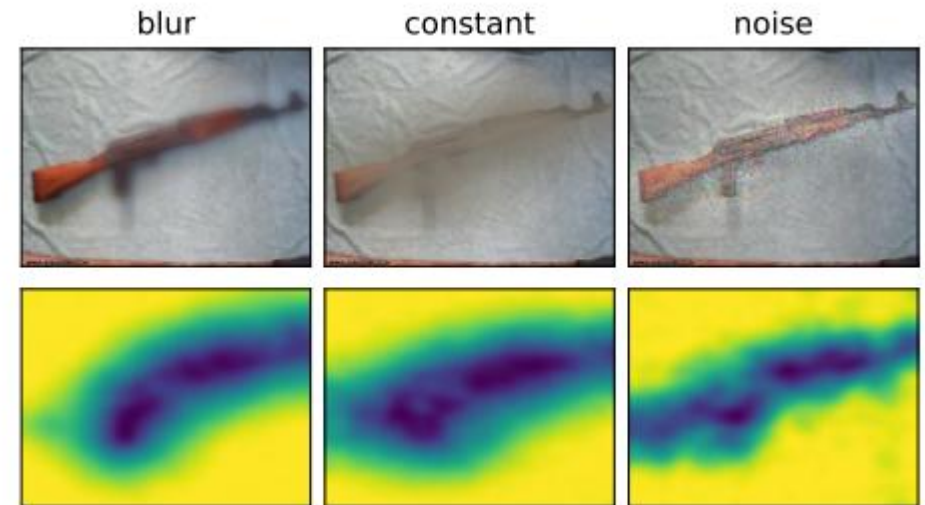
Perturbation-based attributions



$$A_i^P(x) = F(x) - F(x^i)$$

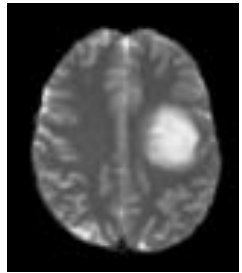
Limitations

- Perturbations out of distribution
- Low resolution

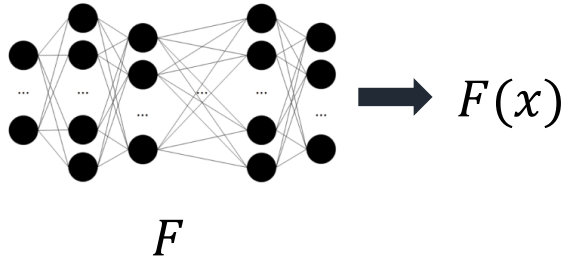


Counterfactual explanations

Trained network

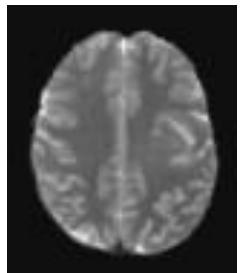


x

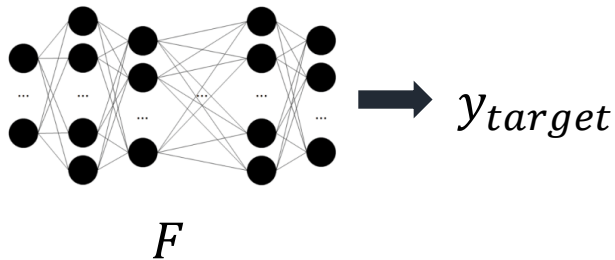


$$\min_{\alpha} \|F(x - \alpha) - y_{target}\| + \lambda \|\alpha\|$$

Minimum changes to the input to change the output

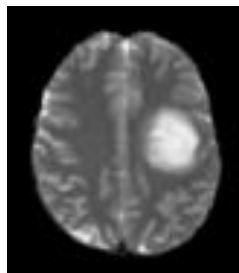


$x - \alpha$

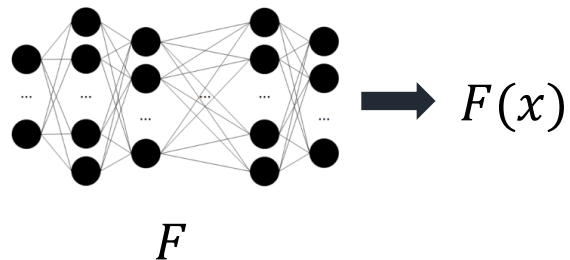


Counterfactual explanations

Trained network

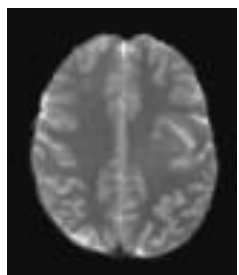


x

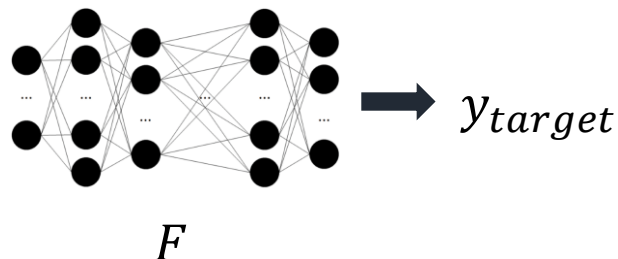


$$\min_{\alpha} \|F(x - \alpha) - y_{target}\| + \lambda \|\alpha\|$$

Minimum changes to the input to change the output



$x - \alpha$



Problem: not easily readable

Counterfactual explanations: not readable

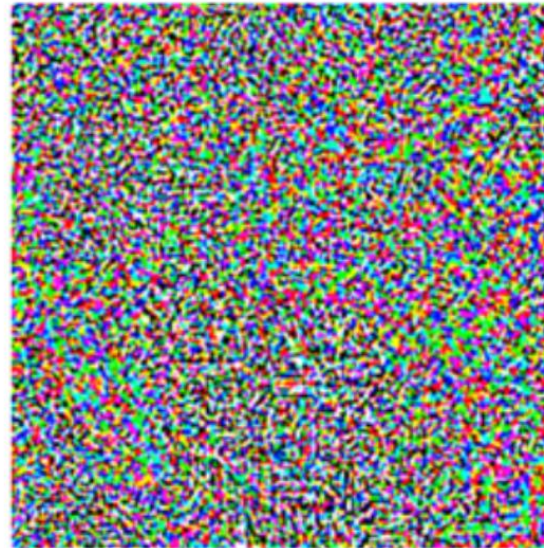


x

“panda”

57.7% confidence

$+ .007 \times$



$\text{sign}(\nabla_x J(\theta, x, y))$

“nematode”

8.2% confidence

$=$



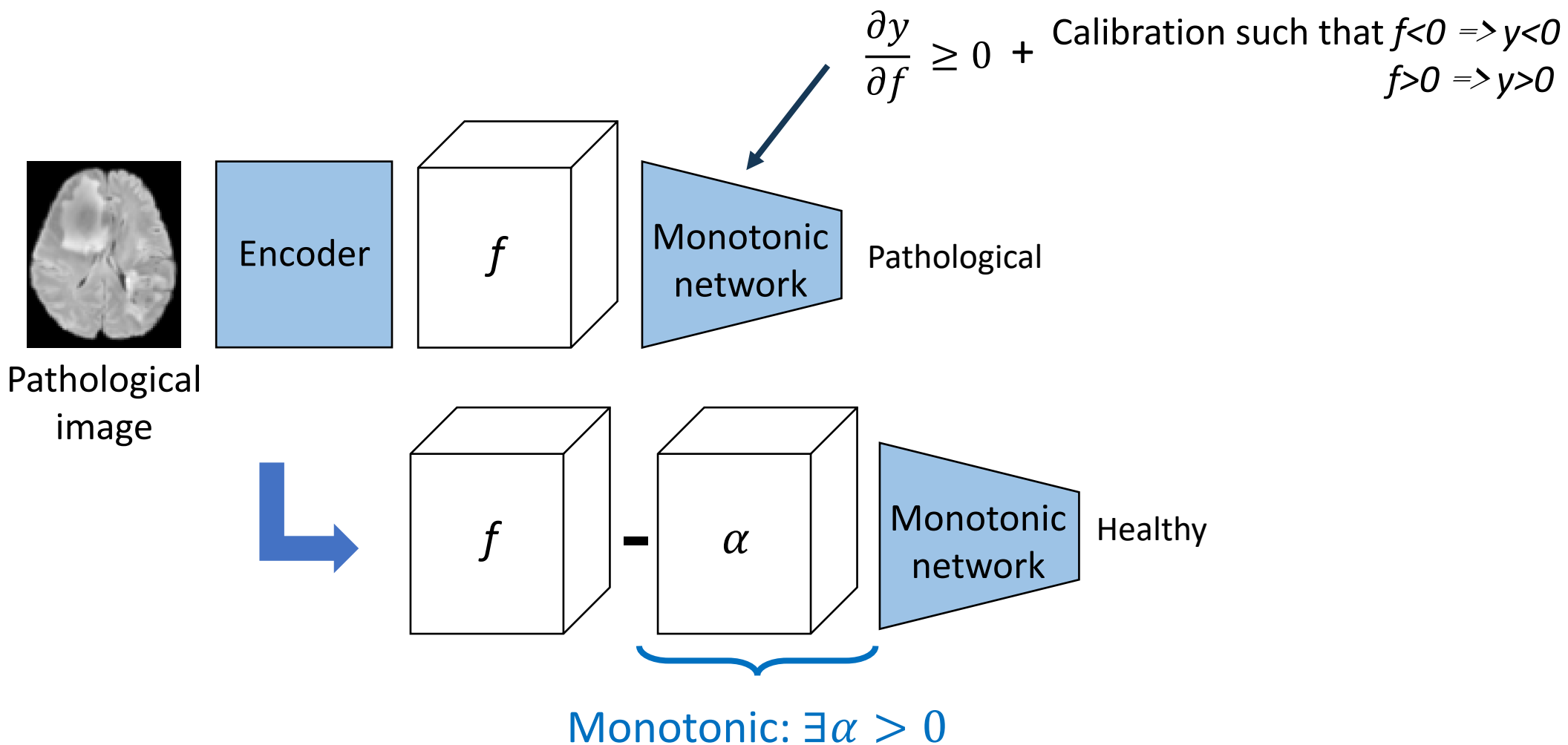
$x + \epsilon \text{sign}(\nabla_x J(\theta, x, y))$

“gibbon”

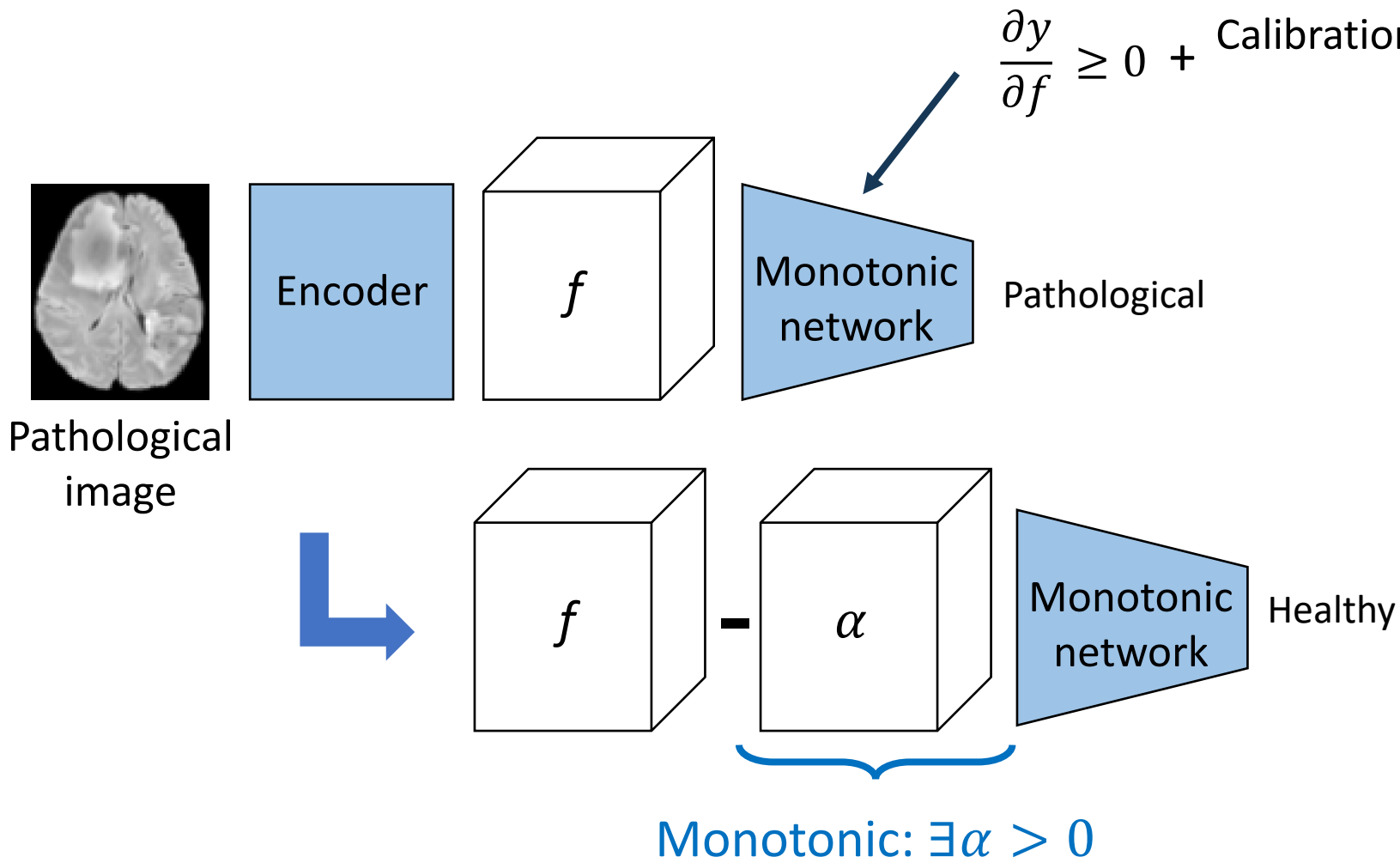
99.3 % confidence

Goodfellow et al. 2014

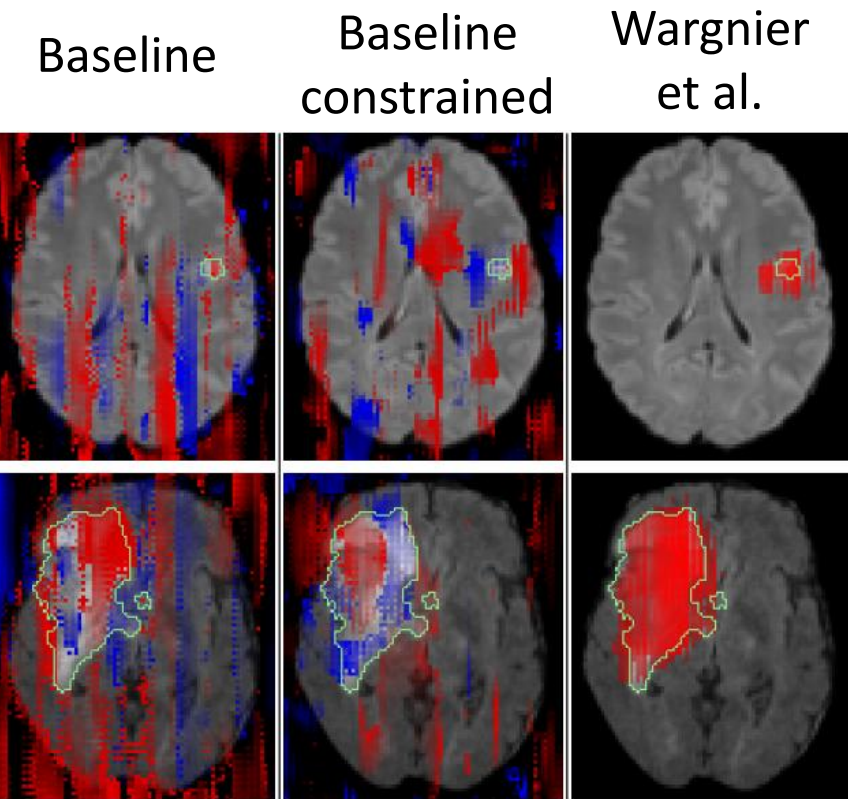
Improve counterfactual explanations



Improve counterfactual explanations

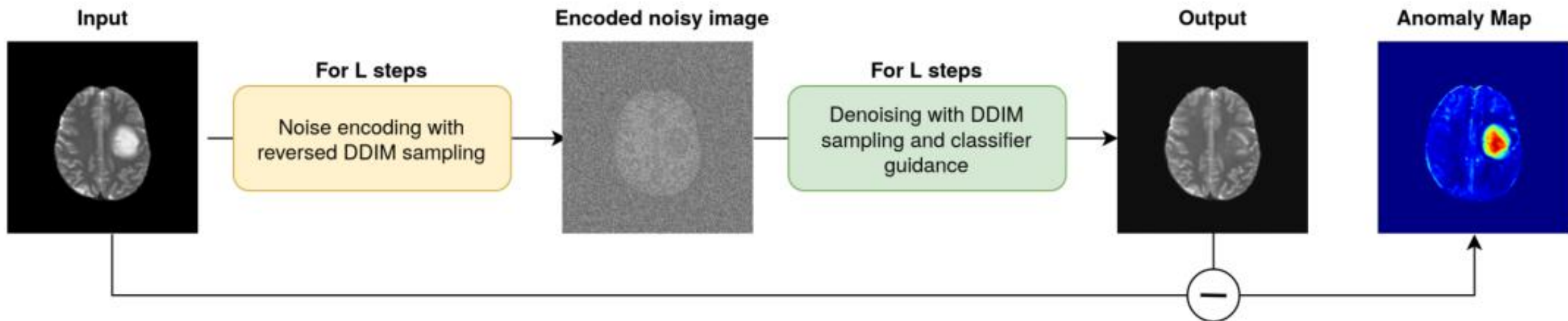


Counterfactual difference α :



Wagnier-Dauchelle et al. Pattern Recognition. 2025

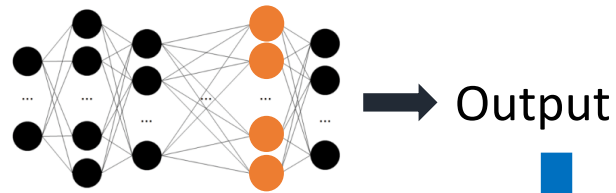
Counterfactual with generative models



Generally: trained for explainability



Input



Semantic concepts

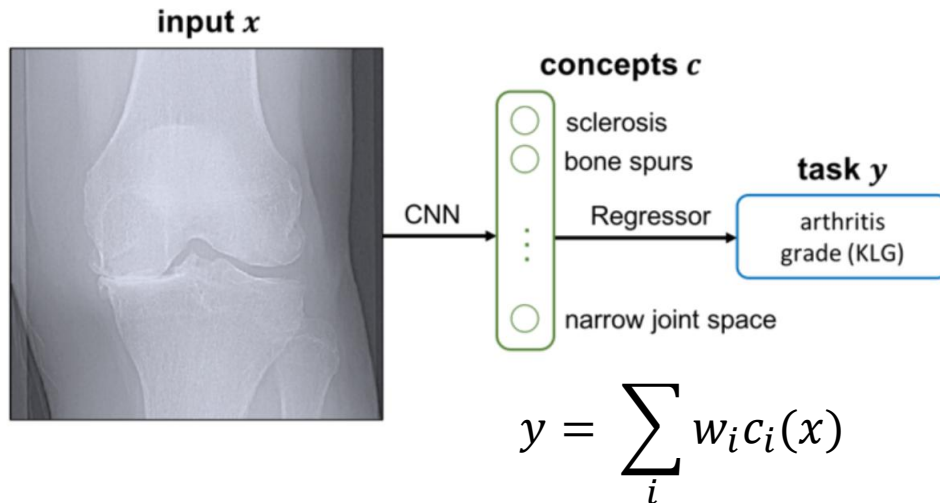
Output

List of concepts

Which semantic concepts are used by the network to take its decision?

- Concept bottleneck
- Prototypes

Pre-defined concepts:



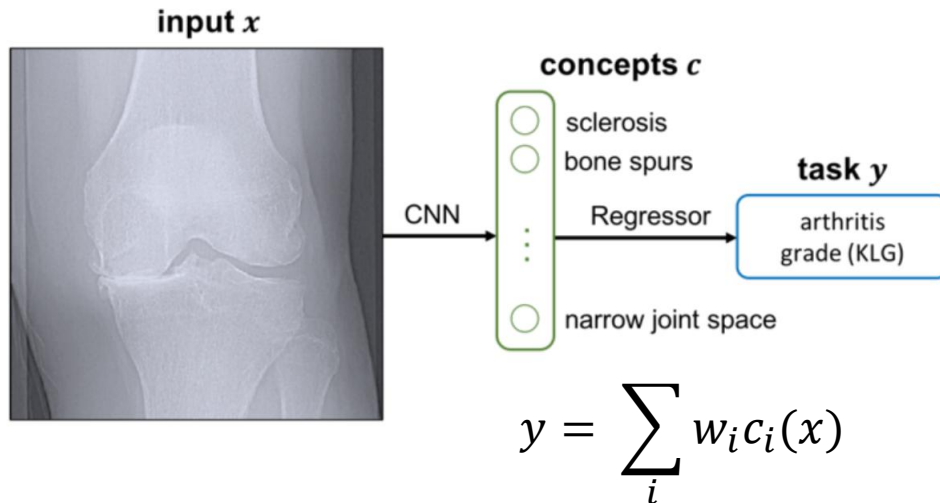
Training loss: classification + right concepts

Limitations

- Annotations

Concept bottleneck models

Pre-defined concepts:

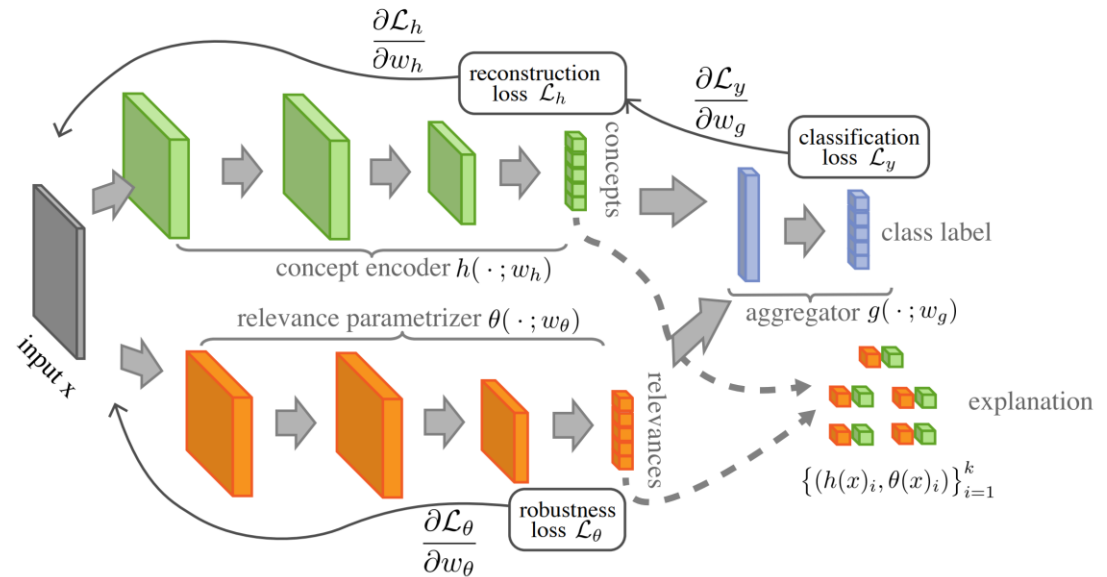


Training loss: classification + right concepts

Limitations

- Annotations

Learned concepts:

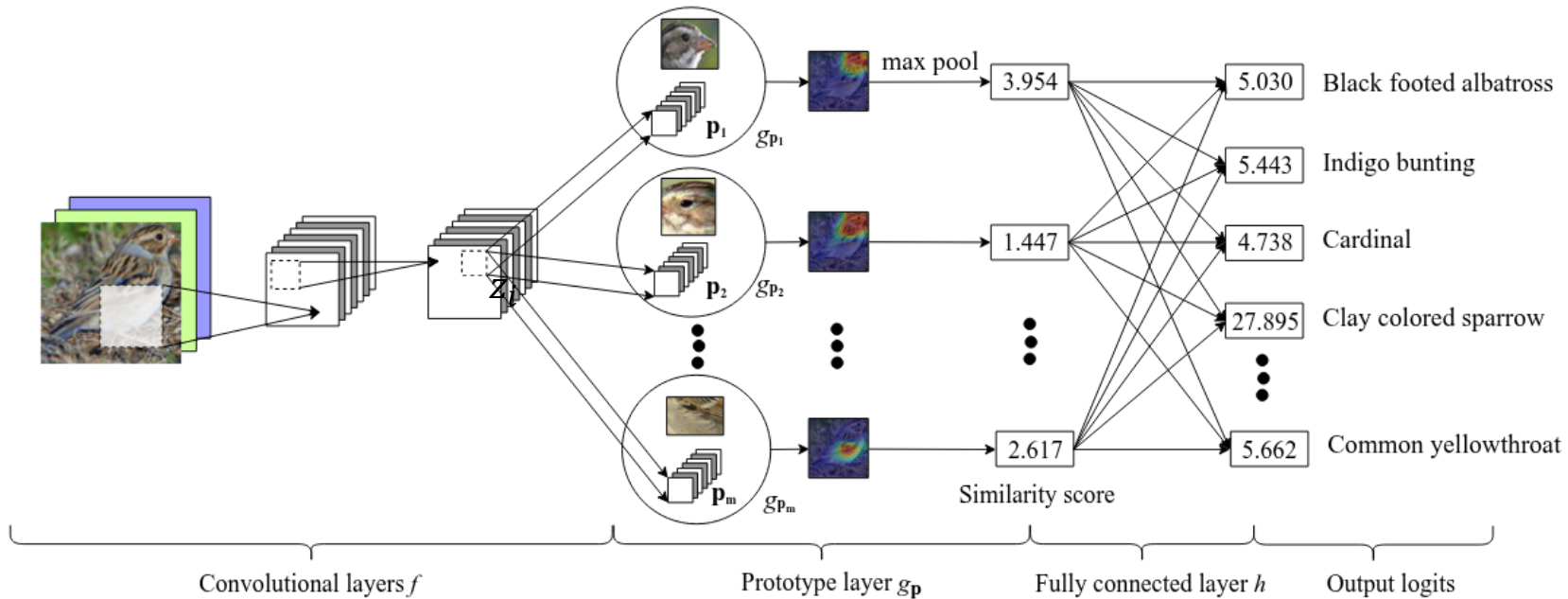


Training losses: - classification
- reconstruction
- regularizations: concepts sparsity
+ local linearity between concepts and relevances

Limitations

- Concepts are abstract

Prototypes models



Fixed number of prototypes per class

Clustering:

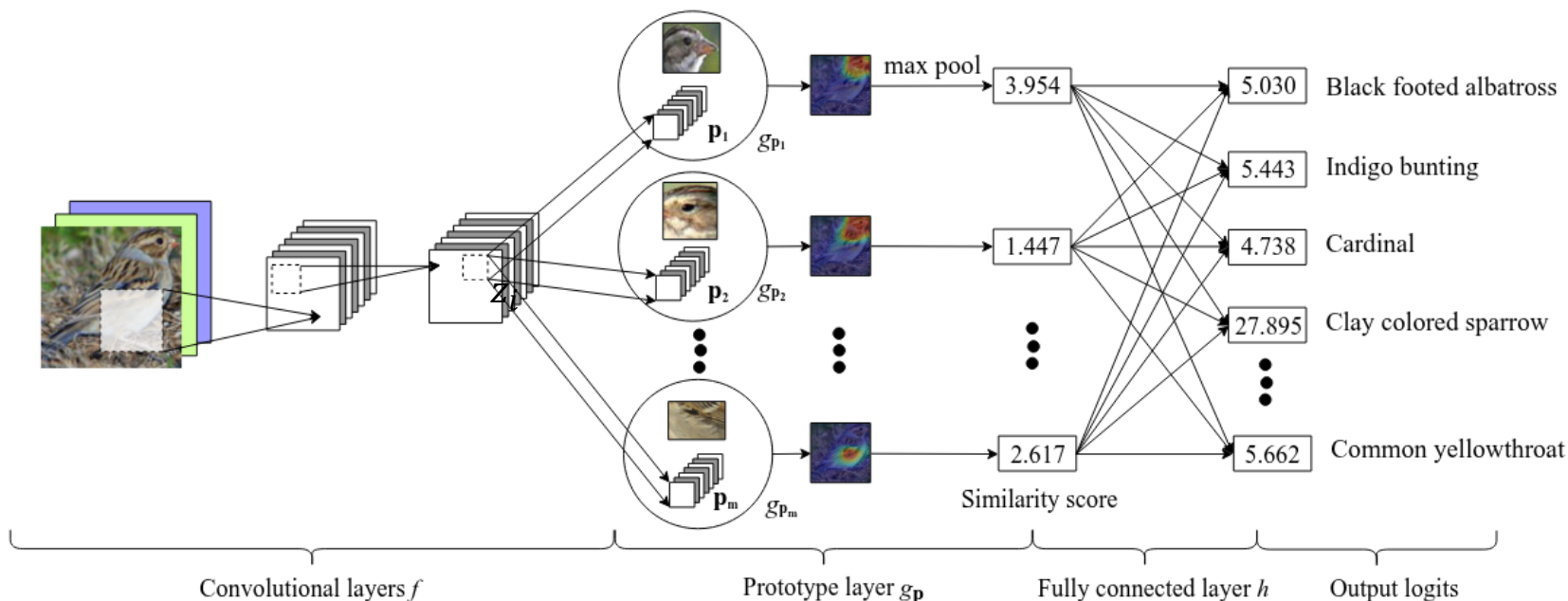
- If z_i and p_i are from the same class
-> closed in the latent space
- If z_i and p_i are from different classes
-> far away in the latent space

**From latent space to image domain:
Decoder**

Projection:

Prototypes from the database

Prototypes models



Fixed number of prototypes per class

Clustering:

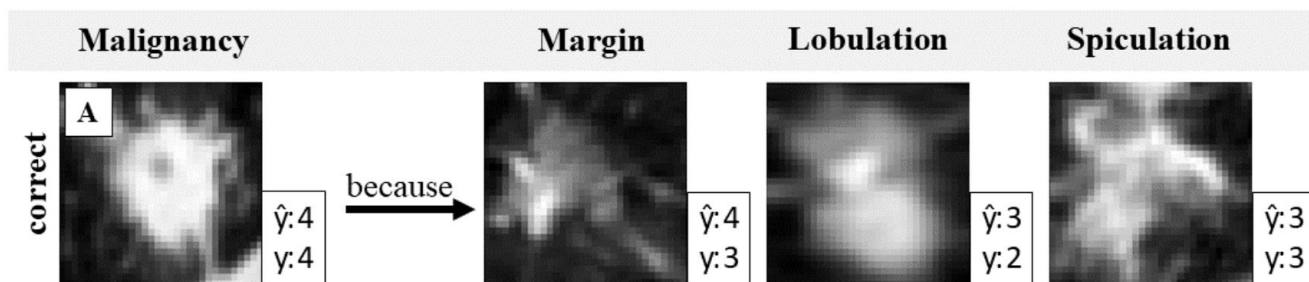
- If z_i and p_i are from the same class
-> closed in the latent space
- If z_i and p_i are from different classes
-> far away in the latent space

From latent space to image domain:
Decoder

Projection:

Prototypes from the database

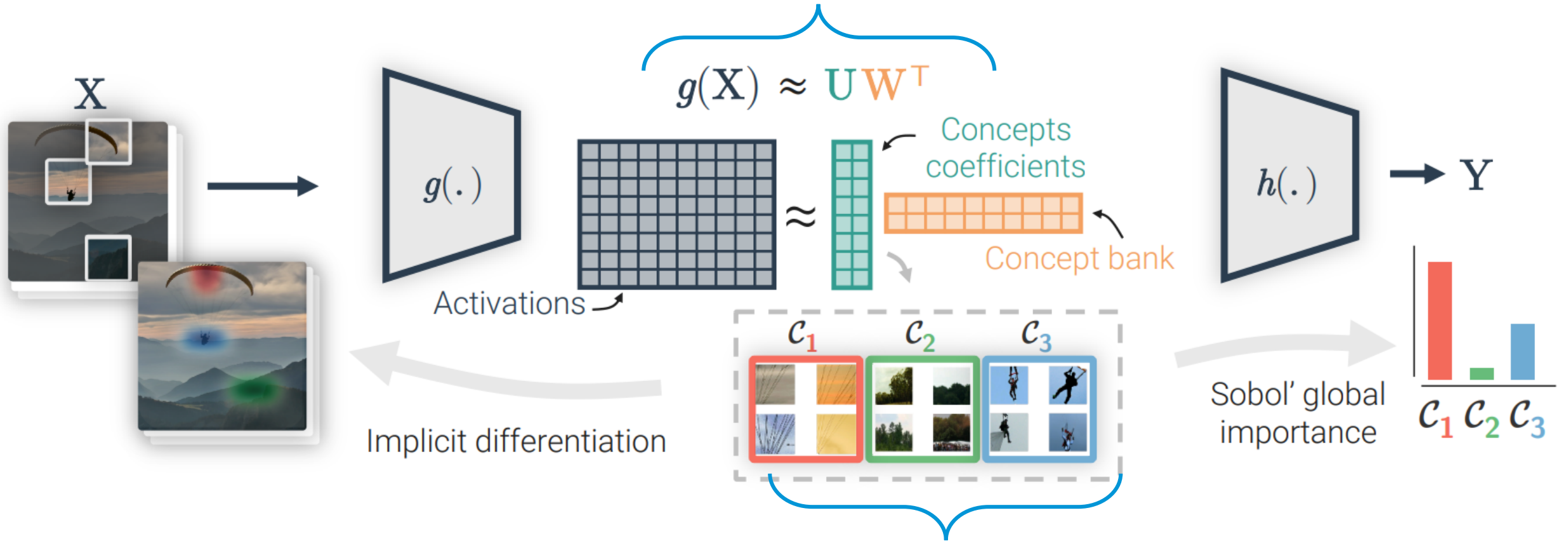
Concepts and prototypes:



Chen et al. Neurips, 2019 Gallée et al. MICCAI, 2023

CRAFT: mix WHERE and WHAT

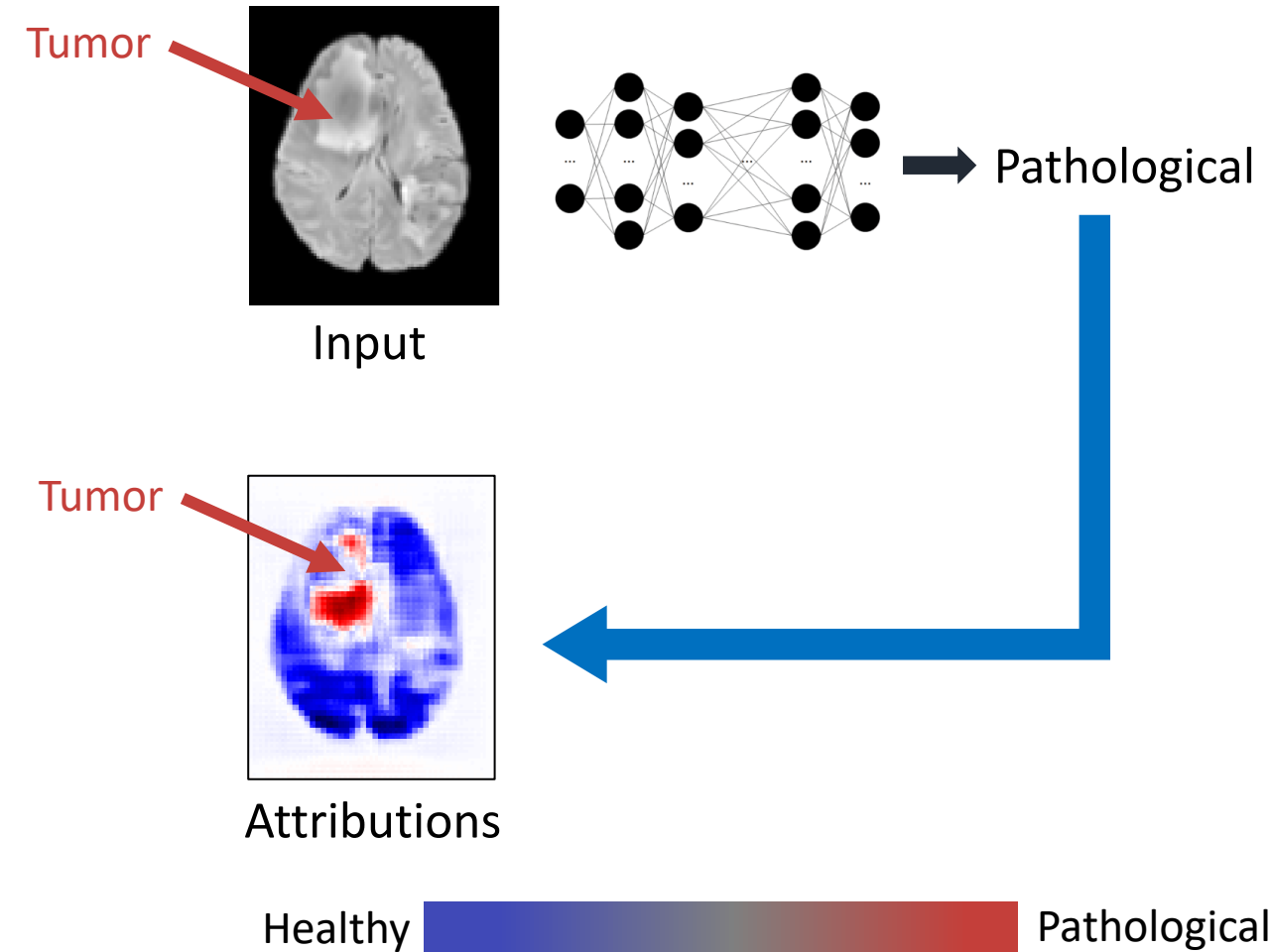
Non-negative Matrix Factorization



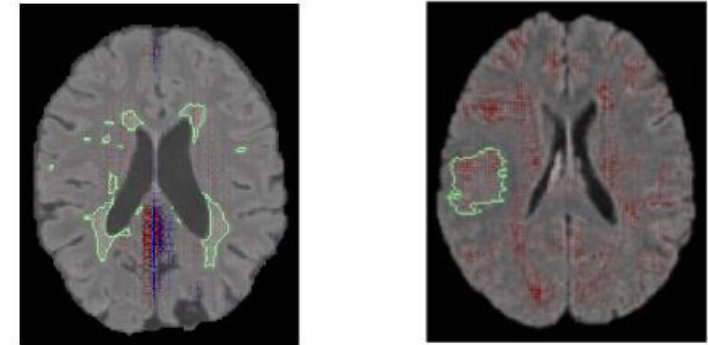
Recursively: sub-concepts with previous layer

Explanation is not sufficient

What we want:



In reality:

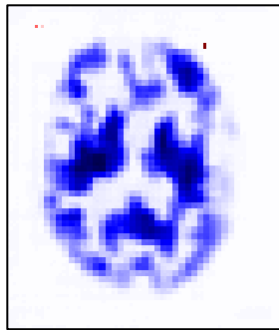
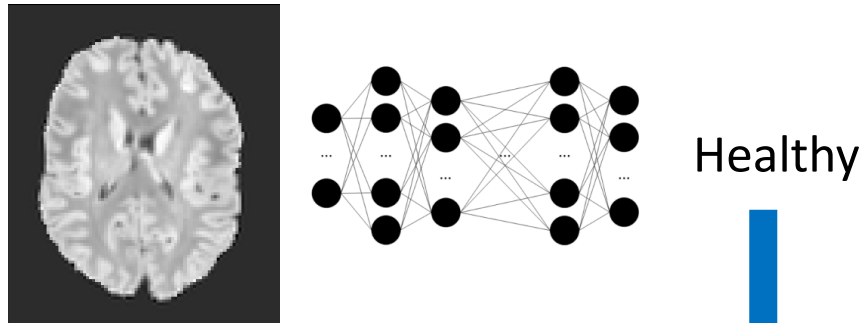


Attributions not focused on relevant areas: shortcuts



Constraints for interpretability

Unsupervised constraint during training:



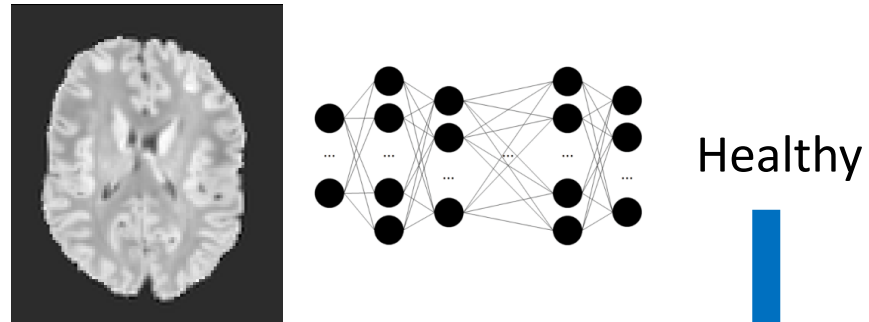
Attributions < 0

Healthy  Pathological

Wagnier-Dauchelle et al. IEEE TMI, 2023

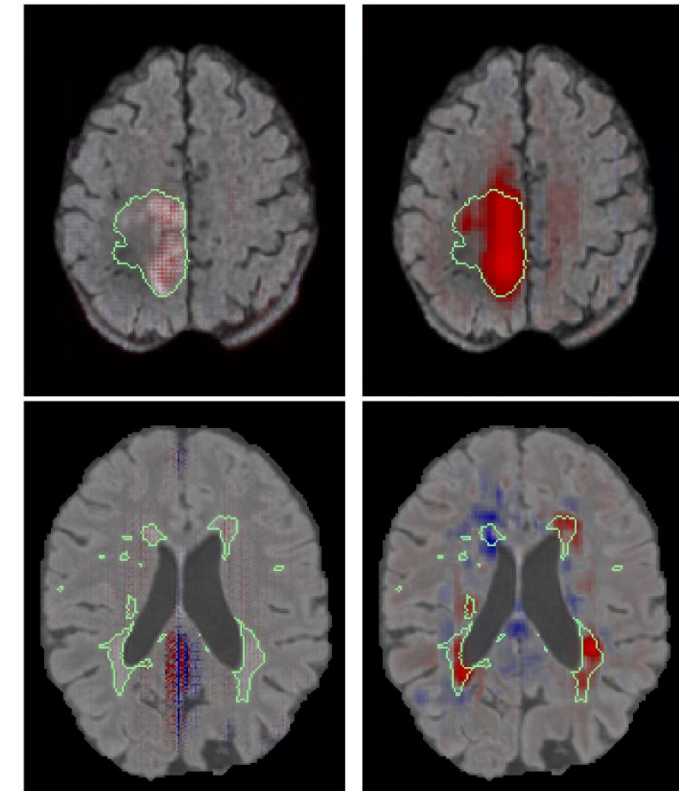
Constraints for interpretability

Unsupervised constraint during training:



Attributions < 0

Healthy  Pathological



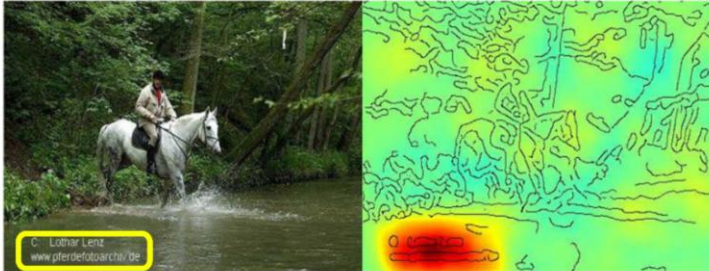
Naive

Constrained

Wagnier-Dauchelle et al. IEEE TMI, 2023

Confounders

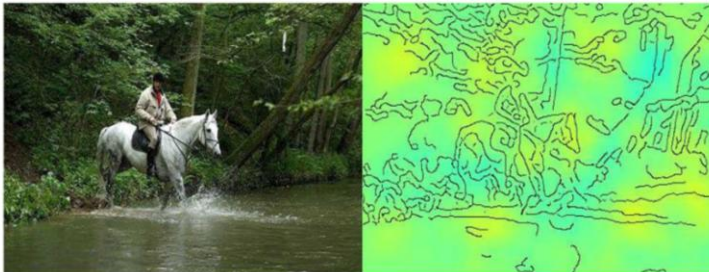
Horse-picture from Pascal VOC data set



Source tag
present



Classified
as horse



No source
tag present



Not classified
as horse

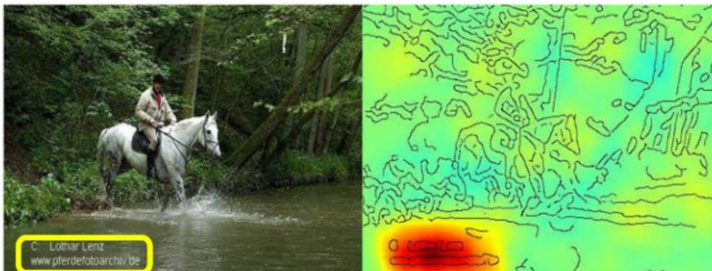
Artificial picture of a car



Lapuschkin et al. Nature communications, 2019

Confounders

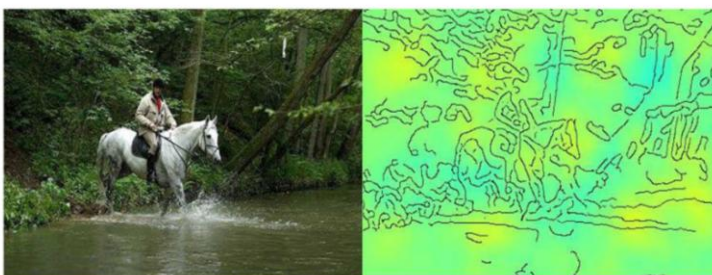
Horse-picture from Pascal VOC data set



Source tag
present



Classified
as horse



No source
tag present

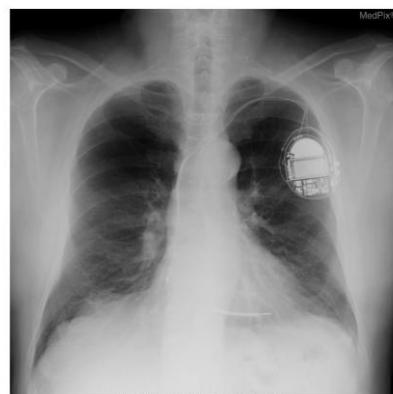


Not classified
as horse

Artificial picture of a car



In medical imaging:



Pacemaker in
chest X-ray



Rule in skin
lesion photo

Lapuschkin et al. Nature communications, 2019

Brain shape as a confounder

Is it possible to distinguish
the two databases using
only the brain shape?



Classifieur

Acc >> 0.5 ?

Brain shape as a confounder

Is it possible to distinguish
the two databases using
only the brain shape?



Classifieur

Acc >> 0.5 ?

Classification task				TNR	TPR	BA
Healthy vs Pathological	IXI	vs	BraTS	1.00	1.00	1.00 ✓
	IXI	vs	OFSEP	0.54	0.70	0.62 ✓
Healthy vs Healthy	IXI	vs	HCP	1.00	1.00	1.00 ✓
	IXI	vs	IBC/kirby/MPI	1.00	0.86	0.93 ✓
	HCP	vs	IBC/kirby/MPI	0.99	0.95	0.97 ✓
Healthy vs Pathological	ADNI-CN	vs	ADNI-AD	0.35	0.74	0.55 ✗

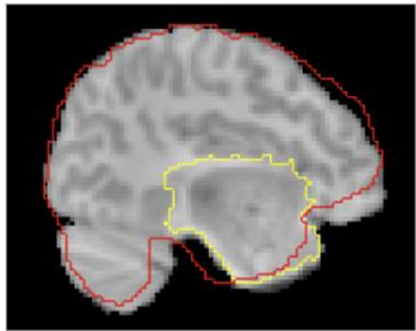
Age matching

Intra dataset

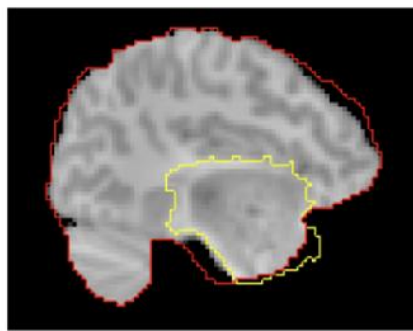
Wagnier-Dauchelle et al. MIDL, 2024

Brain shape as a confounder

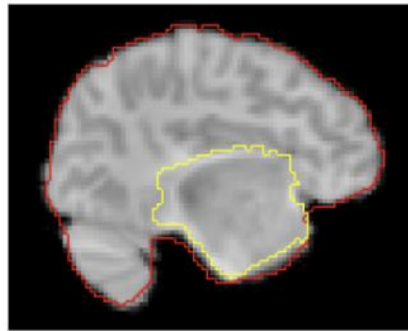
During inference, if the image is modified so that the brain shape is in the distribution of the opposite class, can we still distinguish the two databases just as well ?



(a) Original



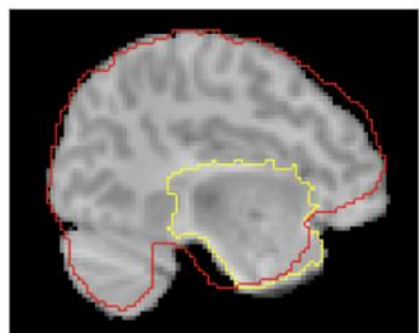
(b) Cropped



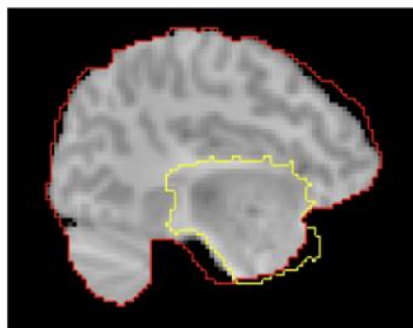
(c) Registered

Brain shape as a confounder

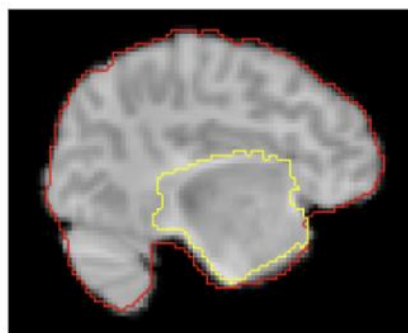
During inference, if the image is modified so that the brain shape is in the distribution of the opposite class, can we still distinguish the two databases just as well ?



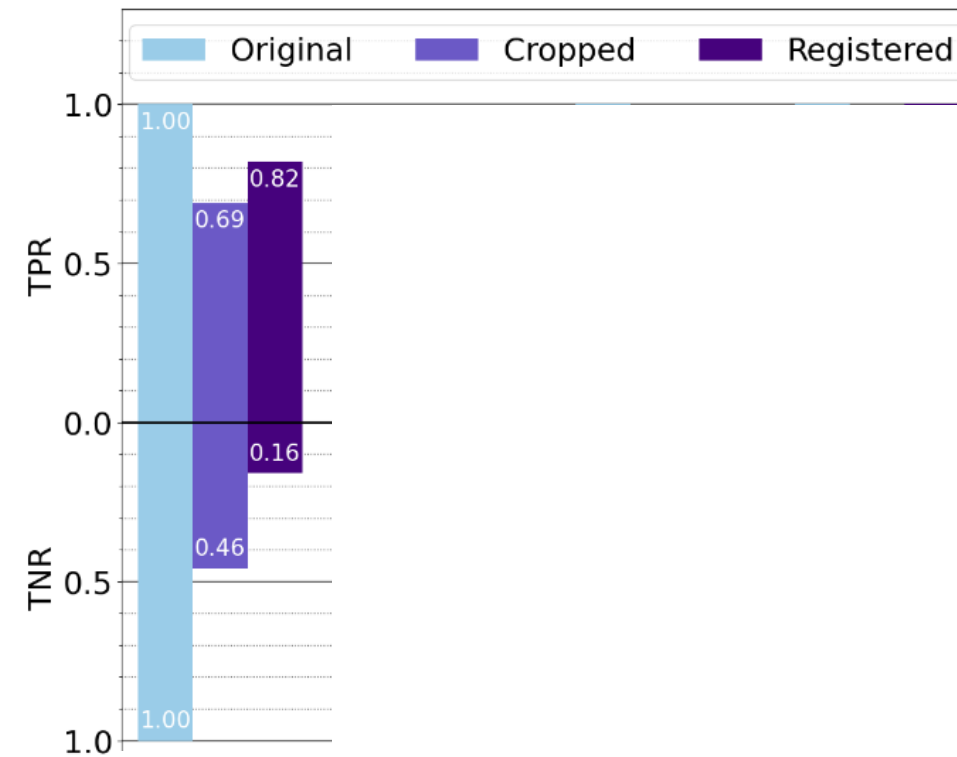
(a) Original



(b) Cropped



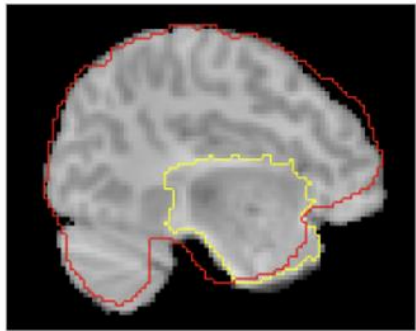
(c) Registered



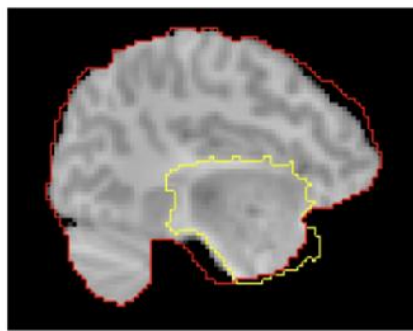
(a) IXI vs BraTS

Brain shape as a confounder

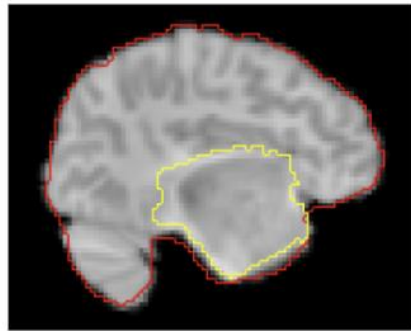
During inference, if the image is modified so that the brain shape is in the distribution of the opposite class, can we still distinguish the two databases just as well ?



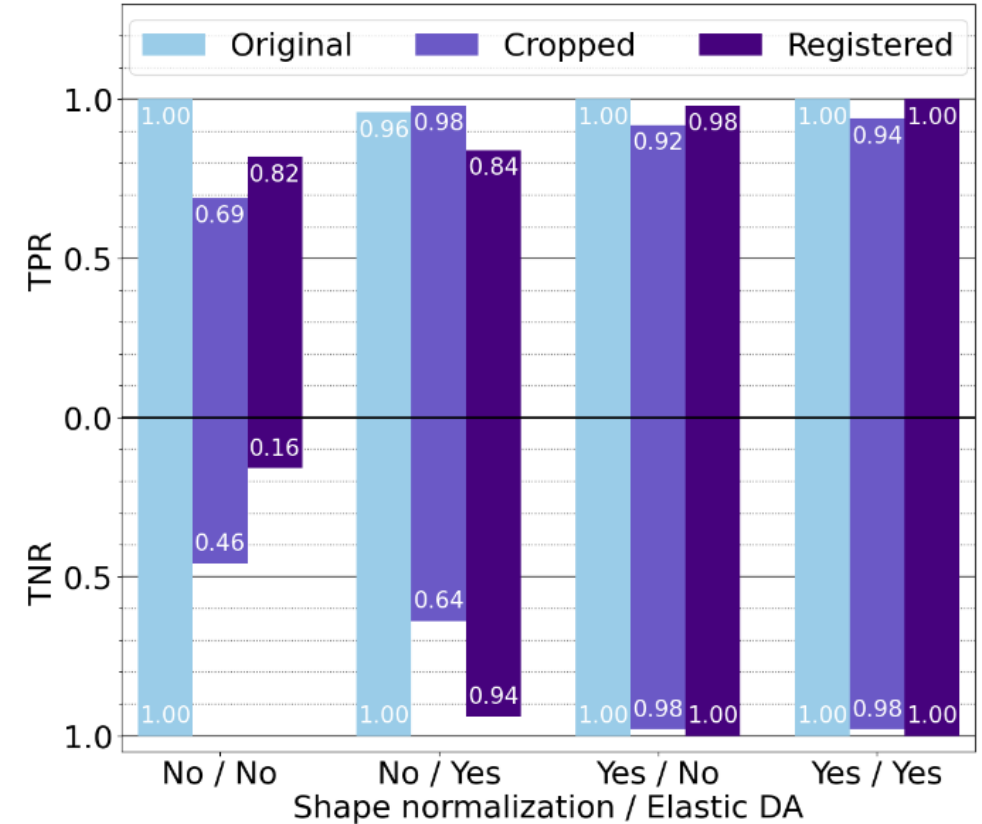
(a) Original



(b) Cropped



(c) Registered

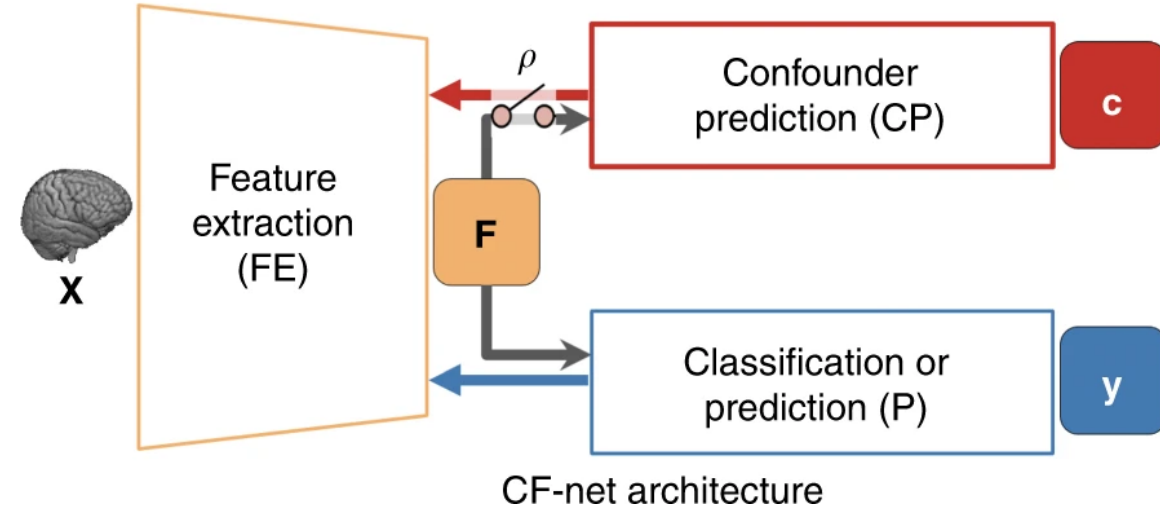


(a) IXI vs BraTS

Solution: brain shape normalization

Wagnier-Dauchelle et al. MIDL, 2024

Confounder free training

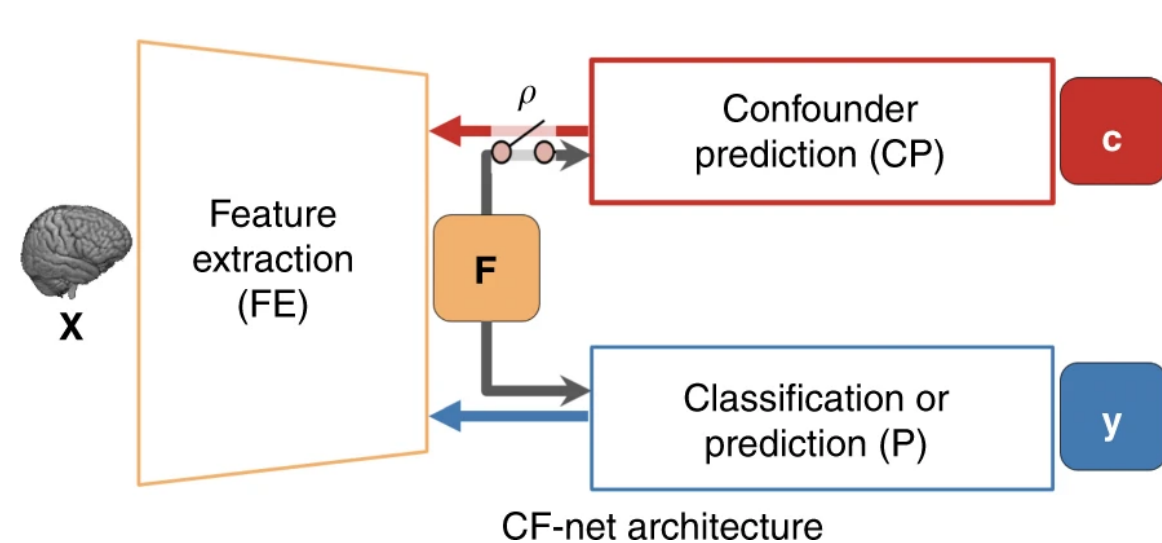


Training strategy:

- Update FE and P to predict y
- Update CP on control only to predict c
→ Avoid confounding of y in CP
- Adversarial update FE to fool CP
→ F cannot predict c

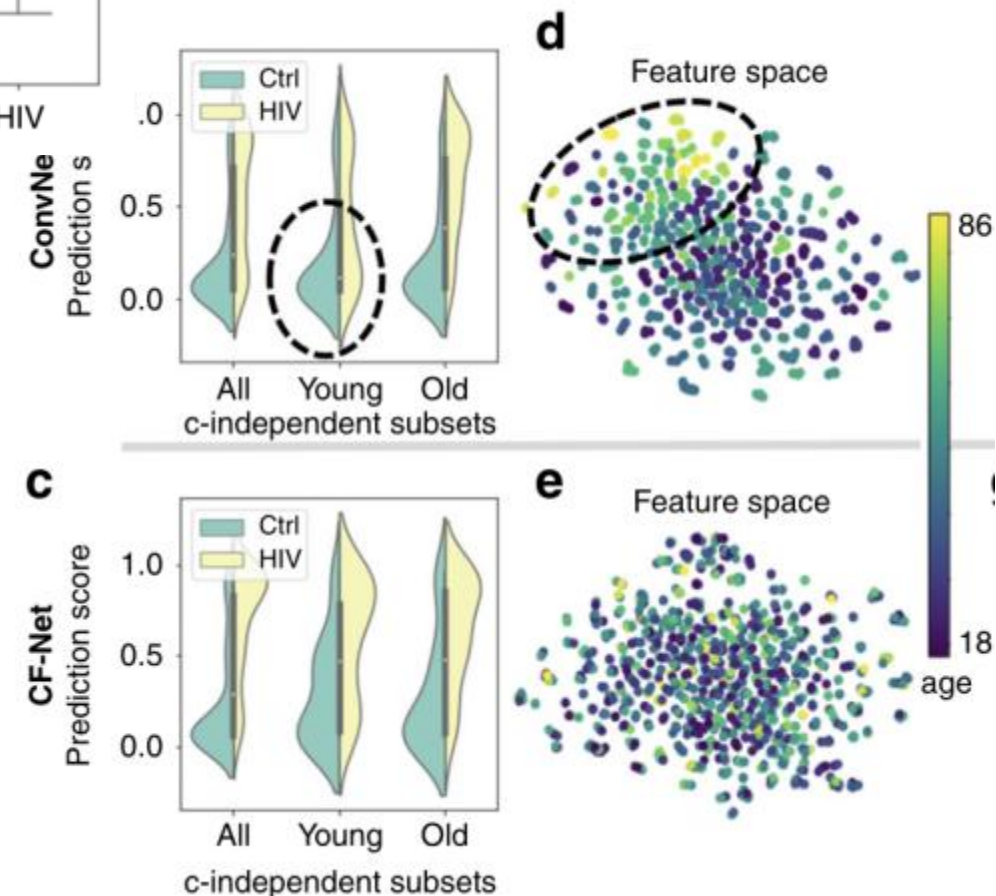
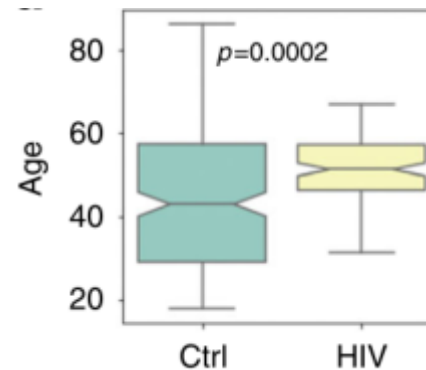
Zhao et al. Nature communications, 2020

Counfounder free training



Training strategy:

- Update FE and P to predict y
- Update CP on control only to predict c
→ Avoid confounding of y in CP
- Adversarial update FE to fool CP
→ F cannot predict c



Thanks